

HAYAT MEDICAL COLLEGE

Histology and Embryology Test II for Year I Medical Students

Time allowed: 2:30 hours.

Name: _____

Date: November 28, 2013

I. HISTOLOGY

MCQ: For each of the following Multiple Choice Questions choose the one BEST appropriate answer and mark 'X' in the appropriate box located against the corresponding question number and letter in the Separate Answer Sheet provided.

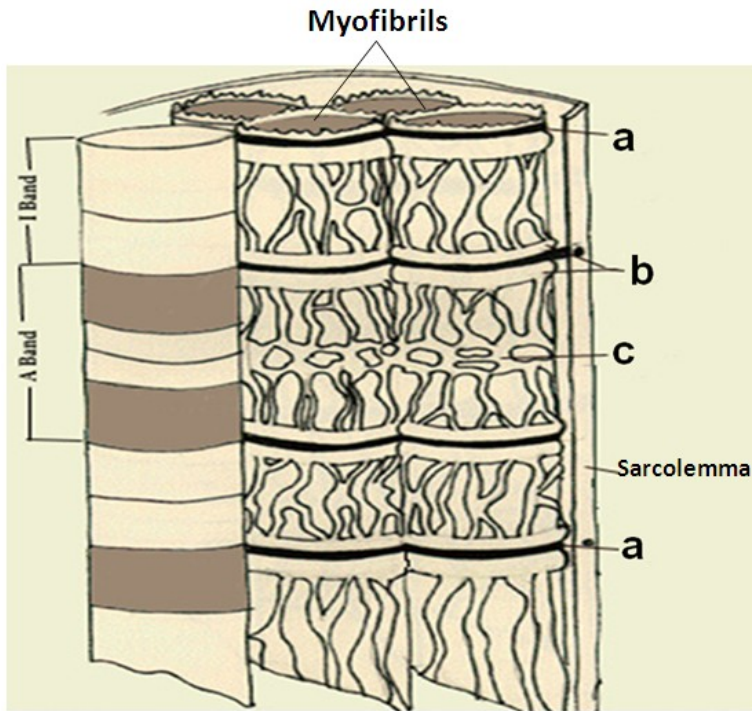
This examination is consisted of **12 pages of question papers and an answer sheet.**

Please write your name on both the front pages of the examination paper and the answer sheet.

- _____ 1. The sarcoplasmic reticulum of **skeletal muscle** cells are made by:
- a) Golgi apparatus b) Polyribosomes c) Rough endoplasmic reticulum.
d) Smooth endoplasmic reticulum. e) Mitochondria.
- _____ 2. Regarding the **striated muscle cells**,
- a) The striations of the light microscopy are manifested because of extensive plasma membrane folding.
b) The location of triads in the myofibrils of skeletal muscle cells is identical in all animals.
c) Both skeletal and cardiac muscle cells show regenerative capacity from the surrounding pericytes if damaged.
d) Intercalated discs of cardiac muscle are made by cell junctions.
e) All skeletal muscles are found attached to bones.
- _____ 3. Region of **cell bodies in the central nervous system** could make all of the following, except:
- a) Gray matter b) Column c) Layer d) Nuclei e) None of the above
- _____ 4. **Nissl bodies** of neurons represent:
- a) Golgi bodies b) Lysosomes c) Microfilaments d) RER e) SER
- _____ 5. **Neurofibrils** are found in which region of the neurone?
- a) Dendrites b) Axon c) Perikarya d) a & b e) All of the above
- _____ 6. Which of the following statements is correct regarding **myelination in the peripheral nervous system**:
- a) Outer lamellae end nearest to the node of Ranvier in comparison to the inner lamellae.

- b) The first layer of the myelin sheath known as inner collar of the Schwann cell contains the nucleus.
- c) One Schwann cell makes more than one segment of myelin sheath for same or different axon(s).
- d) Myelin sheath in the peripheral nervous system has fewer Schmidt-Lanterman clefts than that of the CNS.
- e) Myelin is made mainly by protein as neurokeratin with a smaller amount of phospholipid.

7-10. Answer questions 7-10 from the figure below representing a muscle cell.



- _____ 7. Which type of muscle cell does the figure represent?
- a) Skeletal muscle b) Cardiac muscle c) Smooth muscle d) a & b e) All
- _____ 8. The regions indicated by the letter “a” are:
- a) Terminal cisternae b) Transverse tubule c) Sarcotubules d) Intercalated duct e) Diad
- _____ 9. The regions indicated by the letter “b” are:
- a) Terminal cisternae b) Transverse tubule c) Sarcotubules d) Endomysium e) Diad
- _____ 10. The region indicated by the letter “c” is:
- a) Terminal cisternae b) Transverse tubule c) Sarcotubules d) Mitochondria e) Diad
- _____ 11. Which of the following statements is not true regarding the supporting cells in the **nervous tissue**:
- a) Satellite cells surround the perikarya of the spinal ganglia in a stratified cuboidal fashion.
 - b) Oligodendrocytes are found in both gray and white matters of the central nervous system.
 - c) Microglia are the smallest in size among the neuroglial cells.
 - d) Astrocytes serve in conducting metabolites between blood vessels and neurones.
 - e) Ependymal cells line both the ventricles of the brain as well as the central canal of the spinal cord.

_____ 12. Which of the following **neuroglia** belong to the **mononucleated phagocytic cells**?
a) Astrocytes b) Oligodendrocytes c) Microglia d) Ependyma e) a & b

_____ 13. **Erythrocytes**

- a) Stain acidophilic b) Are biconcave c) Have rigid cell membrane
d) Have centrally placed heterochromatic nucleus e) Are more in number in females than males.

_____ 14. The normal value for **leukocyte count** is

- a) 200,000 – 400,000/ μ l b) 4,100,000 – 6,000,000 / μ l c) 5,000 – 12,000/ μ l
d) 35,000 - 50,000/ μ l e) 2,000,000-5,000,000/ μ l

_____ 15. Which type of **leukocytes** are least in number in the circulating blood?

- a) Basophils b) Eosinophils c) Monocytes d) Neutrophil e) Lymphocytes

_____ 16. Which **leukocytes** increase most in number during bacterial infection?

- a) Basophils b) Eosinophils c) Monocytes d) Neutrophil e) Lymphocytes

_____ 17. At which stage of **hematopoiesis** do morphological characteristics differentiate for the first time, suggesting the mature cell types they will become?

- a) Multipotential stem cells b) Precursor cells c) Progenitor cells
d) Pluripotential stem cells e) None of the above

_____ 18. Stage of **granulopoiesis** with twisted or curved nucleus showing uniform width is:

- a) Metamyelocytes b) Myelocytes c) Promyelocytes d) Myeloblast e) Band stage

_____ 19. Which of the following statements is not correct regarding the arteries?

- a) Elastic arteries do not have smooth myocytes in their tunica media.
b) Tunica media of muscular arteries have elastic and reticular fibres among the smooth myocytes.
c) Tunica adventitia of muscular arteries have collagen and elastic fibres.
d) Most of the named arteries are muscular arteries. e) Aorta is an example of elastic artery.

_____ 20. Which of the following statements is not correct regarding the capillaries?

- a) Continuous capillaries are lined by nonfenestrated endothelium joined by tight junctions.
b) Fenestrated capillaries are lined by endothelium showing several fenestrations, but have continuous basal lamina. c) Hematopoietic organs contain sinusoidal capillaries.
d) Capillaries have thin subendothelial layer. e) All of the above.

_____ 21. Which of the following statements is not correct regarding the veins?

- a) Are more in number than the arteries. b) Have valves which are not found in the arteries.
c) Have thinner wall and larger lumen when compared with the arteries.
d) Have prominent internal elastic lamina. e) Contain larger volume of blood than the arteries.

_____ 22. Which of the following is relatively well developed in large veins as compared to those of arteries?

- a) Tunica intima b) Tunica media c) Tunica adventitia d) All of the above e) None of the above

_____ 23. With regard to impulse generating and conducting system of the heart:

- a) Cells of the sinuatrial node are larger in size when compared to those of the ordinary cardiac myocytes.
- b) Cells of the sinuatrial node have more myofibrils than those of the ordinary cardiac myocytes.
- c) Purkinje cells, when seen under light microscope following H & E staining, appear as large cells with clear zone surrounding centrally placed nucleus.
- d) Sinuatrial node is located in the left atrium.
- e) All are correct.

_____ 24. Regarding the heart:

- a) Myocardium is thicker in the atria than in the ventricles.
- b) Some cardiac myocytes act as endocrine cells.
- c) Is lined internally by stratified squamous epithelium.
- d) Purkinje fibres are found in the epicardium.
- e) Myocardium is made by both cardiac and smooth muscle cells.

_____ 25. Which of the following statements is true regarding the **lymphoid nodules**?

- a) Mainly contain T lymphocytes.
- b) Have connective tissue capsule.
- c) Could be found freely in mucosa associated lymphoid tissue in several regions.
- d) Primary lymphoid nodules show germinative region in the centre.
- e) All of the above.

_____ 26. Which of the following is **not true** about **thymus**?

- a) Has six types of epithelial reticular cells.
- b) Attains to its maximum size at puberty.
- c) Contain no B lymphocyte.
- d) Has both primary and secondary lymphoid nodules in its cortex.
- e) Has Hassall's corpuscles preferentially located in the medulla.

_____ 27. Which of the following structures **does not** make part of **blood-thymus barrier**?

- a) Pericytes
- b) Endothelium
- c) Type I epithelial reticular cells
- d) Basal laminae
- e) Smooth myocytes.

_____ 28. Which of the following statements is **not true** about epithelial reticular cells in the **thymus**?

- a) Stain darker than the lymphocytes in haematoxylin and eosin stained preparations.
- b) Are stellate cells with processes.
- c) Isolate cortex from connective tissue and blood vessels.
- d) Separate cortex from medulla.
- e) All of the above.

_____ 29. Which of the following are **not** normally present in the **subcapsular sinus of a lymph node**?

- a) Macrophages
- b) Reticular cells
- c) Lymphocytes
- d) Reticular fibres
- e) Erythrocytes

_____ 30. Which of the following is **not true** about a **lymph node**?

- a) Afferent lymphatic vessels enter through the hilum.
- b) Efferent lymphatic vessels take away newly formed lymphocytes.
- c) Lymphocytes leave the high endothelial venules in the paracortex.
- d) Has both B and T lymphocytes.
- e) Intermediate lymph sinuses are found in the paracortex.

_____ 31. Which of the following statements is **not true** about the **spleen**?

- a) Is the largest lymphoid organ
- b) Filters blood.
- c) Has both open and closed blood circulation.

d) Has capsule and trabeculae which contain smooth myocytes.

e) Has no B lymphocytes.

_____ 32. Which of the following **lymphoid organs** is involved in **filtering lymph**?

a) Appendix

b) Palatine tonsil

c) Spleen

d) Thymus

e) Lymph node

_____ 33. With regard to the **oral cavity**, which statement is **true**?

a. Sebaceous and eccrine sweat glands are prominent in the transitional [vermillion] zone of a lip.

b. The internal region of the lip is covered by a stratified squamous keratinized epithelium.

c. In the submucosa of the cheeks are minor salivary glands and occasional sebaceous glands (Fordyce spots).

d. Pseudostratified columnar ciliated epithelium covers the oral surface of the soft palate.

e. The palatal raphe is a median fold that joins the internal aspect of each lip to the gum.

_____ 34. Concerning the histology of the **teeth**, which statement is **true**?

a. The highly calcified matrix of the enamel is composed of type III collagen fibrils.

b. The organic matrix of enamel is secreted by odontoblasts.

c. Dentin is a calcified connective tissue that is harder than bone.

d. Ameloblasts are tall, columnar epithelial cells that line the pulp chamber.

e. Like bone, cementum is perforated by Sharpey's fibres and neurovascular elements.

_____ 35. With regard to the **tongue**, which statement is **false**?

a. The filiform papillae are the smallest and most abundant papillae on the dorsum of the tongue.

b. Filiform papillae are the only papillae on the tongue which do not contain taste buds.

c. Foliate papillae are well developed in man and are seen on the edges of the tongue arranged in a "V" shape.

d. Taste buds are found on circumvallate, fungiform, and foliate papillae.

e. Fungiform papillae are mushroom shaped papillae on the tongue which contain taste buds.

_____ 36. Which organ has a **mucosa lined by nonkeratinized stratified squamous epithelium**?

a. Esophagus

b. Stomach

c. Small intestine

d. Large intestine

e. Rectum

_____ 37. Concerning the **stomach**, which statement is **true**?

a. It is specifically responsible for the absorption of vitamin B₁₂ and the reabsorption of conjugated bile salts.

b. Goblet cells are more abundant to produce mucus as well as appreciable amounts of lysozyme.

d. Hydrochloric acid and intrinsic factors are secreted by enteroendocrine cells of cardia and pylorus.

d. Parietal cells also called oxyntic cells contain mucinogen granules and pepsinogen.

e. The folds of the empty stomach are called rugae.

_____ 38. With regard to the **small intestine**, which statement is **true**?

a. The muscularis externa in the upper third of duodenum and jejunum is composed of striated muscle.

b. Crypts of Lieberkühn are lymphatic capillaries within a villus of the small intestine.

c. Microvilli that are seen on the epithelial cells in the small intestine form the brush border or striated border.

d. The plicae circulares are projections with a core of mucosa. e. Villi are finger like projections of submucosa.

_____ 39. With regard to the **large intestine**, which statement is **false**?

a. The Paneth cells of the submucosal glands house bacteria that produce Vitamin B₁₂ and Vitamin K.

b. Lacks villi and contains a disproportionately large number of goblet cells.

c. The lamina propria has many macrophages, plasma cells, eosinophils, and lymphoid nodules.

d. The muscularis externa has three thick bands of muscle called the tenia coli.

e. The large intestine is shorter than the small intestine but has a wide lumen.

_____ 40. With regard to the **pancreas**, which statement is **false**?

- a. It is a lobulated retroperitoneal gland with mixed exocrine-endocrine components.
- b. The exocrine portion is a compound tubuloacinar gland, similar in structure to the parotid gland.
- c. The myoepithelial cells of the striated ducts deliver bicarbonate ions into the primary secretion.
- d. Islets of Langerhans are incompletely delimited from the acini by Schwann cells.
- e. Islet cells are compact mass of epithelial cells not cords or plates.

_____ 41. Which is not the **function of the liver**?

- a. Detoxification/removal of drugs and alcohol.
- b. Fetal RBC production
- c. Storage of glycogen, fat soluble vitamins, iron and cholesterol.
- d. Production of plasma proteins and lipoproteins.
- e. Storage of calcium and bile.

_____ 42. Concerning the **Kupffer cells**, which statement is **false**?

- a. are found on the luminal surface of the endothelial cells, within the sinusoids, but lack basal lamina.
- b. metabolize aged erythrocytes and they digest hemoglobin.
- c. uptake, store, and release of retinoids.
- d. are derived from monocytes and they destroy bacteria that enter the portal blood through the large intestine.
- e. Most of them are located in the periportal region of the liver lobule, where they are very active in phagocytosis.

_____ 43. Concerning the GI tract, select the statement which is not correct.

- a. The mucosa is the innermost layer of the GI tract.
- b. Meissner's plexus is located in the submucosa.
- c. The adventitia consists of only a single layer of epithelial cells and underlying connective tissue.
- d. The muscularis externa consists of Auerbach's plexus.
- e. Brunner's glands in submucosa of the duodenum produce mucin containing alkaline fluid and the polypeptide hormone urogastrone.

_____ 42. With regard to the salivary glands, which statement is false?

- a. Parotid Gland is a branched acinar gland.
- b. secretory portion of the parotid gland is composed exclusively of serous cells.
- c. The branched tubuloacinar submandibular gland possess mucous cells as the main component.
- d. The short intercalated ducts of the submandibular gland are lined with cuboidal epithelium.
- e. Sublingual gland is a branched tubuloacinar gland formed of serous and mucous cells.

_____ 46. Where is thick skin found?

- a. Over the knee
- b. Sole of the feet
- c. Breast
- d. Lips
- e. All of the above

_____ 47. With regard to the epidermis, which statement is false?

- a. The cells in the stratum granulosum contain keratohyaline granules.
- b. The stratum lucidum is present only in thick skin.
- c. Langerhans cells are macrophages seen in the skin.
- d. The keratinocyte is the most abundant cell in the epidermis.
- e. Fibroblasts form stratified squamous epithelium of the epidermis.

_____ 48. What type of glands are the ceruminous glands?

- a. Sebaceous glands
- b. Eccrine sweat gland
- c. Endocrine gland
- d. Apocrine sweat gland
- e. Oil gland

_____ 48. The most common cells in dermis are:

- a. keratinocytes (or "prickle cells").
- b. melanocytes.
- c. Langerhans cells.

d. Fibroblasts.

e. Merkel cells.

_____ 32. What is the innervation of an eccrine sweat gland?

a. Cholinergic; parasympathetic

c. Adrenergic; parasympathetic

d. Adrenergic; sympathetic

b. Cholinergic; sympathetic

e. Cholinergic; motor

_____ 32. Which of the following is NOT considered an epidermal appendage?

a. Sweat gland b. Hair _____ c. Hypodermis d. Nails e. Sebaceous glands

The hypodermis is the loose connective tissue layer underneath the dermis.

_____ 32. Which of the following is NOT a function of skin

a. Vitamin D production

b. Protection from water loss

c. Sensory reception

d. Heat regulation

e. All are functions of skin

_____ 32. Which of the following is not a function of the kidney?

a. Erythropoietin production

b. Vitamin D modification

c. Acid base balance

d. Aldosterone production

e. Renin production

_____ 10. What vessels arise from the efferent arterioles?

a. Afferent arteriole b. Efferent arteriole c. Capillary tuft

d. Peritubular capillaries

e. Vasa recta

_____ 11. Select the *correct statement* pertaining the *morphology of the kidney*.

a. The distal convoluted tubules exhibit brush borders and apical canaliculi.

b. Macula densa cells are located in the urinary pole of the nephrons.

c. Renin is secreted from the visceral layer of Bowman's capsule referred to as area cribrosa.

d. Interstitial cells in the renal medulla secrete erythropoietin through the filtration slits.

e. The renal pyramid and its associated cortex is referred to as the renal lobe.

_____ 12. In the *urinary system*, which statement is *false*?

a. The ureter is lined by transitional epithelium and it does not contain lymphatics.

b. Polyploidy and binucleated cells are common among cells of the transitional epithelium.

c. Epithelial cells lining the thin limb of the loop of Henle are simple squamous.

d. Mesangial cells are modified smooth muscle cells responsible for the secretion of rennin

e. Aldosterone acts on the distal convoluted duct.

_____ 32.

_____ 32.

_____ 32.

_____ 32.

_____ 32.

1. Mucosa of the gastrointestinal tract contains the gut associated lymphatic tissue.

Brunner's glands are glands in the submucosa of the small intestine.

Goblet cells are most numerous in the large intestine

10. Where is mesothelium found?

- a. Mucosa b. Sub mucosa c. Muscularis externa d. Serosa e. Adventitia

The space between the liver sinusoids and the hepatocytes is called the space of Disse.

8. What is the functional unit of the liver?

- a. Lobule b. Portal triad c. Central vein d. Hepatocyte e. Sinusoids

4. In what structure does blood run through between the hepatocytes?

- a. Hepatic artery b. Portal triad c. Central vein d. Portal vein e. Sinusoids

6. What layer is NOT found in the gallbladder?

- a. Mucosa b. Muscularis mucosa c. Muscularis d. Adventita e. Serosa

The mucosa of the gallbladder is made of simple columnar epithelium.

Once bile is produced by the liver, it is stored in the gallbladder.

The small intestine is the primary site for absorption of nutrients.

The masticatory mucosa is located on the hard palate and gingiva.

7. Where are apocrine sweat glands NOT found?

- a. Areola b. External genitalia c. Posterior neck d. Axilla e. Circumanal region

8. What type of tissue makes up the dermis of the skin?

- a. Mucous connective tissue b. Mesenchyme c. Loose irregular connective tissue
d. Dense irregular connective tissue e. Dense regular connective tissue

1. The projections of dermis upward, indenting the bottom of the epidermis, are called:

- a. arrector pili.
b. epidermal appendages.
c. Meissner's corpuscles.
d. dermal papillae.
e. sweat ducts.

———34. **Sebaceous gland** is a type of:

- a) Compound tubular gland. b) Compound alveolar gland. c) Simple straight tubular gland.
d) Simple branched alveolar gland. e) Simple coiled tubular gland.

———39. Which of the following statements is **true** about the **lip**?

- a) The inner aspect has sebaceous glands. b) The outer aspect has apocrine sweat glands.
c) The transitional border has abundant salivary glands.
d) The inner surface is lined by stratified columnar epithelium. e) The core contains skeletal muscles.

————40. Which of the following statements is **true** regarding the **teeth**?

- a) Odontoblastic processes branch and taper as they approach dentinoenamel junction.
- b) Dentin is not present in the deciduous teeth.
- c) Enamel has more collagen fibres than dentin.
- d) Enamel is found overlying dentin in both crown and root portion of the teeth.
- e) Enamel develops from mesoderm, while both dentin and cementum develop from overlying ectoderm.

————41. **Muscularis mucosae of esophagus:**

- a) Is made by inner oblique, middle circular and outer longitudinal layers.
- b) Is made by inner circular and outer longitudinal layers.
- c) Is found only as longitudinal layer.
- d) Is entirely of skeletal muscle type.
- e) None of the above.

————42. Which of the following statements is **true** about the **stomach**?

- a) Gastric glands of cardiac region are compound branched tubuloalveolar in type
- b) Gastric glands of pyloric region have longer glands than the pit with roughly 4:1 ratio
- c) Has mucosa lined with stratified squamous nonkeratinized epithelium.
- d) The most inner layer of muscularis externa is arranged as inner circular layer
- e) Cells of the gastric glands that produce HCl are eosinophilic in staining

————43. Which of the following statements is **true** about the **small intestine**?

- a) Has plicae circularis made by transverse folds of the circular muscle layer of muscularis externa
- b) Has microvilli made by projections from absorptive cells
- c) Has its most outer layer made by partly serosa and partly adventitia
- d) Has its muscularis mucosae made by skeletal muscle
- e) Has plicae circularis that increase the surface area more than microvilli or villi

————44. Which of the following is **correct** about the **histology of pancreas**?

- a) Although is a mixed exocrine and endocrine gland, the majority of the secretory cells are exocrine cells.
- b) The exocrine cells are arranged as simple tubular type.
- c) The islets of Langerhans are made by larger and more eosinophilic cells than the exocrine cells.
- d) The exocrine portions have more capillaries than the endocrine portion.
- e) The B cells of islets of Langerhans are distinguished under light microscope by being larger and more basophilic than the other cells of the islets.

II. EMBRYOLOGY

MCQ: For each of the following Multiple Choice Questions choose the one BEST appropriate answer and mark 'X' in the appropriate box located against the corresponding question number and letter in the Separate Answer Sheet provided.

- _____ 1. Which of the following statements is false regarding the human **placenta**?
a) Maternal portion is formed by deciduas parietals. b) Fetal portion is formed by chorion frondosum.
c) Cotyledons are seen from the maternal aspect. d) Fetal portion is smooth. e) Is hemochorial.
- _____ 2. Which of the following statements is **false** regarding the human **placenta**?
a) Floating villi are involved in the exchange of substances between the maternal and fetal blood more than the stem villi. b) The stem villi serve to anchor the placenta onto the maternal tissue.
c) The fetal blood freely circulates from umbilical arteries into the intervillous space and then back via umbilical vein. d) The fetal border is covered by amnion.
e) At full term its attachment extends to cover about 30% of the endometrium.
- _____ 3. Which of the following is **false** concerning the **amniotic fluid**?
a) Normally amounts to 400 - 500 ml during the last trimester. b) About 99% of it is made by water.
c) Is produced primarily from the maternal interstitial fluid. d) Contains fetal feces and urine
e) Is taken into both the gastrointestinal and respiratory tracts of the fetus.
- _____ 4. When **placenta** is developed at the lower segment, near the cervix of the uterus, it is known as:
a) Placenta previa b) Placenta succenturiata c) Battledore
placenta
d) Placenta accrete e) Placenta membranacea

_____ 5. When **monozygotic twins** are formed by split at morula stage, the twins will have

- a) Same chorionic sac and placenta, but separate amniotic cavity
- b) Same chorionic sac, but separate placenta and amniotic cavity
- c) Same chorionic sac, placenta and amniotic cavity
- d) Separate chorionic sac, placenta and amniotic cavity
- e) Separate chorionic sac and placenta, but same amniotic cavity

_____ 6. With regard to the **congenital anomalies**, which of the following statements is **false**?

- a) At birth, the incidence of minor congenital anomalies is less than that of major congenital anomalies.
- b) Infants with multiple minor anomalies most frequently have one or more associated major anomalies as well.
- c) Recessively inheritable mutations may cause birth defect while occurring in single (heterozygous) dose.
- d) Monosomy of the sex chromosome is known to cause human birth defect seen at birth.
- e) A similar teratogen may not cause identical congenital anomalies in different genomes.

_____ 7. The causes for most of the currently detected major human **birth defects** at birth are:

- a) Genetic factors
- b) Environmental factors
- c) Combination of genetic and environmental factors
- d) Unknown factors
- e)

_____ 8. Most **birth defects** occur during which of the following periods/stages of the intrauterine life?

- a) First 2 weeks
- b) Embryonic period
- c) Fetal period
- d) Last trimester
- e) Zygote

_____ 9. A condition where there is a decrease or an increase from the normal **diploid number of a specific chromosome** as cause(s) for **birth defect** is known as:

- a) Polyploidy
- b) Hypodiploidy
- c) Aneuploidy
- d) Hyperdiploidy
- e) Ploidy

_____ 10. Which of the following structures develop from the costal processes of the **sclerotomes**?

- a) The transverse processes
- b) Foramina transversaria
- c) Lateral mass or alae of the sacrum
- d) All of the above
- e) None of the above

_____ 11. The cranial half of the **1st cervical sclerotome**:

- a) Gives rise to the upper part of body of the 1st cervical vertebra.
- b) Gives rise to the lower part of body of the 1st cervical vertebra.
- c) Gives rise to the whole body of the 1st cervical vertebra.
- d) Participates in the formation of base of occipital bone.
- e) Disappears

_____ 12. The body of **1st thoracic vertebra** is formed by:

- a) The caudal half of the 8th cervical sclerotome alone.
- b) The cranial half of the 1st thoracic sclerotome alone.
- c) The caudal half of the 1st thoracic sclerotome alone.
- d) Fusion of caudal half of the 1st thoracic with cranial half of the 2nd thoracic sclerotomes.
- e) Fusion of caudal half of the 8th cervical with cranial half of the 1st thoracic sclerotomes.

_____ 13. The spinal nerves, in relation to their corresponding **sclerotomes**, were initially located:

- a) Above their corresponding sclerotomes.
- b) Below their corresponding sclerotomes.
- c) At the level of succeeding sclerotome.
- d) At the level of preceding sclerotome.
- e) At the level of middle of corresponding sclerotomes.

_____ 14. **Bones of the limbs** develop from:

- a) Somatic layer of lateral mesoderm.
- b) Visceral layer of lateral mesoderm.
- c) Sclerotomes.
- d) Intermediate mesoderm.
- e) Neural crest.

_____ 15. Which of the following statements is **correct** with regard to the **rotation of the limbs during the 7th week?**

- a) Lower limb rotates 180° medially, while upper limb rotates 180° laterally.
- b) Lower limb rotates 180° laterally, while upper limb rotates 180° medially.
- c) Lower limb rotates 90° laterally, while upper limb rotates 90° medially.
- d) Lower limb rotates 90° medially, while upper limb rotates 90° laterally.
- e) Lower limb rotates 90° laterally, while upper limb rotates 180° medially.

_____ 16. The dorsal somatic mesodermal condensations in the limb bud give rise to which types of muscles?

- a) Extensors and pronators.
- b) Extensors and abductors.
- c) Extensors and flexors.
- d) Flexors and pronators.
- e) Flexors and abductors.

_____ 17. Which of the following statements is **true** regarding the **hypomeric and epimeric divisions of the myotomes?**

- a) Epimers are larger and located dorsally as compared to the hypomers.
- b) Epimers are smaller and located dorsally as compared to the hypomers.
- c) Epimers are innervated by the dorsal branches of the ventral primary rami of the spinal nerves.
- d) Hypomers give rise to the extensors of the back.
- e) None of the above.

_____ 18. **Cardiac muscles** develop from:

- a) Somatic layer of lateral mesoderm.
- b) Splanchnic layer of lateral mesoderm.
- c) Intermediate mesoderm.
- d) Both splanchnic and somatic mesoderm.
- e) Myotomes.

_____ 19. Which of the following is a condition caused by early **closure of epiphyseal plates of long bones leading to dwarfism?**

- a) Acrania
- b) Acromegally
- c) Achondroplasia
- d) Klippel-Feil syndrome
- e) None of the above

_____ 20. Which of the following **bones** does not develop in the 1st arch?

- a) Incus
- b) Malleus
- c) Maxilla
- d) Mandible
- e) Styloid process of temporal bone

_____ 21. Structure that is **not** developed in the **third branchial arch** includes:

- a) Lesser cornu of hyoid bone
- b) Glossopharyngeal nerve
- c) Common carotid artery
- d) Stylopharyngeus muscle
- e) Lower part of body of hyoid bone

_____ 22. Structures derived from endoderm of the **fourth pharyngeal pouch** include

- a) Thymus gland
- b) Palatine tonsil
- c) Thyroid gland
- d) Parafollicular cells of thyroid gland
- e) Lingual tonsil

_____ 23. The **anterior 2/3rd of the tongue** develops from

- a) Hypobranchial eminence
- b) The median lingual swelling
- c) Cupula
- d) The lateral lingual swellings
- e) Epiglottal swelling

_____ 32. Which of the following is **not** a component of **vernix caseosa**?

- a) Secretions from sebaceous gland
- b) Secretions of apocrine sweat glands
- c) Fallen epidermal cells
- d) Fallen hairs
- e) None of the above

_____ 33. Which of the following structures in the **skin** develops from a different germ layer than the others?

- a) Hairs
- b) Sweat glands
- c) Arrector pili muscle
- d) Nails
- e) Apocrine sweat glands

_____ 79. **Concerning the development of vasculature,**

- a) distal parts of the 3rd pair of aortic arches join the dorsal aortae to form the internal carotid arteries.***
- b) the left 4th aortic arch becomes the proximal part of the left subclavian artery.***
- c) the distal part of the right 6th aortic arch becomes the ductus arteriosus.***
- d) the superior vena cava forms from the right anterior cardinal vein and the right posterior cardinal vein.***
- e) after birth, the proximal parts of the umbilical arteries persist as the superior vesical arteries.***

_____ 81. With regard to the development of the digestive system,

- a) the foregut may give rise to the lower respiratory system.
- b) the midgut may give rise to the descending colon and sigmoid colon.
- c) the hindgut derivatives are supplied by inferior mesenteric artery.
- d) during the rotation of the stomach, the lesser curvature moves to the left and the greater curvature moves to the right.
- e) the dorsal pancreatic bud forms the uncinate process and part of the head of the pancreas.

_____ 82. During the development of the digestive system,

- a) omphalocele is a persistence of the herniation of abdominal contents into the proximal part of the umbilical cord.
- b) the superior two-thirds of the adult anal canal are derived from proctodeum.
- c) the junction of the epithelium is derived from the ectoderm of the proctodeum.
- d) the inferior one-third of the anal canal is supplied with superior rectal artery.
- e) the anal membrane normally breaks down by the end of the second trimester..

_____ 4. Concerning the development of the GIT and associated structures, which statement is true?

- a) Liver develops by fusion of ventral and dorsal hepatic buds.
- b) Stomach rotates 270° counterclockwise, similar to the midgut.
- c) The loop of the midgut has the superior mesenteric artery in its axis.
- d) The hepatocytes of liver are of mesodermal in origin, while the stroma is of endodermal origin.
- e) The exocrine portion of pancreas develops from endoderm of the gut, while the endocrine portion develops from the local mesoderm.

_____ 26. During the development of the digestive system, which statement is true?

- a) omphalocele is a persistence of the herniation of abdominal contents into the proximal part of the umbilical cord.
- b) the superior two-thirds of the adult anal canal are derived from proctodeum.
- c) the midgut may give rise to the descending colon and sigmoid colon.
- d) the dorsal pancreatic bud forms the uncinate process and part of the head of the pancreas.
- e) the anal membrane normally breaks down by the end of the second trimester.

_____ 34. The midgut begins with _____ and ends with _____.

- a) 1/2 jejunum; 2/3 transverse colon b) 1/2 ileum; 2/3 sigmoid colon c) 1/2 jejunum; 2/3 rectum
d) 1/2 duodenum; 2/3 transverse colon e) 1/2 duodenum; 1/3 sigmoid colon

_____ 35. The middle of the duodenum would receive blood from:

- a) Superior mesenteric artery only b) Inferior mesenteric artery only c) Celiac artery only
d) Celiac and inferior mesenteric artery e) Celiac and superior mesenteric artery

_____ 36. What two structures only form on the ventral side of the bile duct?

- a) Liver; Pancreas b) Liver; Gallbladder c) Pancreas; Gallbladder
d) Heart; Liver e) Heart; Pancreas

_____ 37. During week six, the midgut herniates through the umbilical ring, rotating _____ degrees counter clockwise. Near week ten, the midgut returns, rotating _____ degrees for an overall rotation of _____ degrees.

- a) 180; 180; 0 b) 180; 180; 360 c) 90; 180; 90 d) 90; 180; 270
e) 270; 90; 180

_____ 38. A three-week old male infant comes into the Emergency Room for weight loss and acute dehydration. The infant has been projectile vomiting and the condition is worsening. Which of the following is most likely malformation?

- a) Gastric atresia b) Pyloric stenosis c) Annular pancreas
d) Malrotation of the gut e) Intestinal duplication

_____ 39. Meckel's diverticulum is a congenital connection from the umbilicus via a vitelline ligament to

- a) Duodenum b) Jejunum c) Ileum d) Transverse colon e) Sigmoid colon

_____ 40. During the pronephros period (week 3), the first kidney develops in the _____ region. By week 4 this kidney degenerates.

- a) Cervical b) Thoracic c) Lumbar d) Sacral e) Cloaca

_____ 5. With regard to the congenital anomalies of the kidneys and ureters which statement is false?

- a) Accessory renal arteries are end arteries.
b) Bilateral renal agenesis is associated with oligohydramnios.
c) Incontinence is a common complaint resulting from an ectopic ureteric orifice in both sexes.
d) Unilateral renal agenesis is more common in males than females. e) None of the above.

_____ 6. Which of the following is not the adult derivative of the ureteric bud?

- a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule

_____ 7. Urachal cyst is a remnant of:

- a) urogenital sinus b) urogenital ridge c) cloaca d) allantois e) mesonephric duct

_____ 8. The podocytes of Bowman's capsule are derived from:

- a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

_____ 11. During the development of the kidney systems, which statement is true?

- a) The transitional epithelium lining the urethra is derived from neuroectoderm.
- b) The “ascent” of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.
- c) The pronephros are derived from intermediate mesoderm from upper thoracic to upper lumbar segments.
- d) Urine production begins shortly after birth.
- e) At birth, the kidneys have a lobulated appearance.

_____ 17. Concerning the development of the Urinary System, which statement is false?

- a) The kidneys attain their definitive adult position postnatally.
- b) Urinary system begins to develop before the genital system.
- c) The metanephric diverticulum is the primordium of the ureter.
- d) Nephrons are derived from the metanephric mass of intermediate mesoderm.
- e) Differentiation of the nephrons depends upon induction by the collecting tubules.

_____ 18. Which of the following is not the adult derivative of the ureteric bud?

- a) renal pelvis
- b) major calyx
- c) minor calyx
- d) collecting duct
- e) connecting tubule

_____ 19. Concerning the development of the urinary bladder and urethra, which statement is false?

- a) The female urethra is derived from endoderm of the urogenital sinus.
- b) In adults, the urachus is represented by the median umbilical ligament.
- c) The urinary bladder becomes a pelvic organ during the 9th week of development.
- d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
- e) Epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.

_____ 41. The ureteric bud gives rise to the _____ and the metanephric mesoderm gives rise to the _____.

- a) Nephrons; Collecting system
- b) Collecting system; Nephrons
- c) Bowman's capsule; Ureter
- d) Loop of Henle; Renal pelvis
- e) Distal/proximal convoluted tubules; Major/minor calyces

_____ 42. The urorectal septum divides the _____ from the _____.

- a) Anorectal canal; Urogenital sinus
- b) Urogenital sinus; Mesonephric duct
- c) Urachus; Allantois
- d) Urinary bladder; Allantois
- e) Mesonephric duct; Ureteric bud

_____ 43. The urachus forms the:

- a) Allantois ligament
- b) Median umbilical ligament
- c) Medial umbilical ligament
- d) Lateral umbilical ligament
- e) Round (umbilical) ligament

_____ 44. The bladder becomes continuous with the lower end of the mesonephric (Wolffian) duct, which

- a) Urethra
- b) Prostate
- c) Trigone
- d) Posterior wall
- e) Anterior wall

appropriate answer and mark 'X' in the appropriate box located against the corresponding question number and letter in the Separate Answer Sheet provided.

_____ 11. The muscular part of interventricular septum is formed by:

- a) Downward extension of the spiral aorticopulmonary septum
- b) Fusion of superior and inferior endocardial cushions of the ventricular wall
- c) Fusion of right and left endocardial cushions of the ventricular wall
- d) Upward growth of a midline septum from the floor of a common ventricle
- e) Expansion of the lateral aspects of primitive ventricle with the midline region remained unexpanded

_____ 12. The common carotid arteries develop from:

- a) Truncus arteriosus
- b) 1st aortic arch arteries
- c) 2nd aortic arch arteries
- d) 3rd aortic arch arteries
- e) Dorsal aortae

_____ 13. In fetal blood circulation oxygenated blood is not contaminated by deoxygenated blood in which of the following sites?

- a) Left atrium
- b) Right atrium
- c) Inferior vena cava
- d) Descending aorta
- e) None of the above

_____ 14. Which of the following veins participates in the formation of the inferior vena cava?

- a) Right vitelline vein
- b) Right posterior cardinal vein
- c) Right umbilical vein
- d) Left umbilical vein
- e) Right posterior cardinal vein

_____ 15. Portal vein is derivative of:

- a) Right vitelline vein
- b) Left vitelline vein
- c) Anastomosis between left and right vitelline veins
- d) Anastomosis between right umbilical and right vitelline veins
- e) Anastomosis between left umbilical and left vitelline veins

_____ 16. After birth umbilical vein becomes:

- a) Medial umbilical ligament
- b) Median umbilical ligament
- c) Lateral umbilical ligament
- d) Ligamentum teres hepatis
- e) Ligamentum venosum

_____ 18. Which of the following statements is correct regarding to the **development of the gut?**

- a) Esophagus was short during early development, but gets elongated towards the end of embryonic period.
- b) The midgut was connected to the yolk sac by the allantois.
- c) The primitive gut tube is in open communication with the amniotic cavity.
- d) The site of junction of foregut and midgut derivatives is the 2nd part of duodenum just proximal to the entrance of common bile duct.
- e) All are correct

_____ 19. Concerning the **development of pancreas,**

- a) The ventral bud arises from the proximal part of hepatic bud.
- b) The ventral bud rotates & comes to lie anterior to the dorsal bud.

- c) The ventral bud arises from the distal part of foregut, while the dorsal bud arises from the proximal part of midgut.
- d) The exocrine portion develops from endoderm, while the endocrine portion develops from local mesoderm.
- e) The ventral bud appears before the appearance of the dorsal bud.

_____ **40.** Concerning the **development of the Urinary System**, which statement is **false**?

- a) Urinary system begins to develop before the genital system
- b) The metanephric diverticulum is the primordium of the ureter.
- c) Nephrons are derived from the metanephric mass of intermediate mesoderm.
- d) Differentiation of the nephrons depends upon induction by the collecting tubules.
- e) The kidneys attain their definitive adult position toward the end of the fetal period.

_____ **41.** Concerning the **development of the urinary bladder and urethra**, which statement is **false**?

- a) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
- b) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.
- c) In adults, the urachus is represented by the median umbilical ligament.
- d) The urinary bladder becomes a pelvic organ during the 9th week of development.
- e) The female urethra is derived from endoderm of the urogenital sinus.

_____ **42.** During the **development of the Urinary System**, which statement is **true**?

- a) Bilateral renal agenesis is associated with oligohydramnios.
- b) The podocytes of Bowman's capsule are derived from endoderm.
- c) Connecting tubule is the adult derivative of the ureteric bud.
- d) The kidneys attain their definitive adult position postnatally.
- e) Urine production begins shortly after birth.

-
-

- 2. 1. The term mucosa (mucous membrane) means:
 - a. a nonkeratinized stratified squamous epithelium.
 - b. an epithelium which contains mucus-secreting goblet cells.
 - c. a body surface onto which mucous glands secrete their product.
 - d. the lining of the GI tract.
 - e. the moist surface of any hollow organ that communicates with the outside world.
- 3.
- 4. 4. Lamina propria typically differs from ordinary loose connective tissue by including:
 - a. abundant adipose tissue.
 - b. mucous glands.
 - c. numerous white blood cells.
 - d. serous glands.
 - e. densely interwoven collagen fibers.
- 5.
- 6. 6. Lamina propria differs from dermis by having:
 - a. more adipocytes.
 - b. more densely packed collagen fibers.

- c. more lymphocytes.
- d. fewer capillaries.
- e. fewer macrophages and mast cells.

7.

8. 7. The epithelium lining the *entire* GI tract is:

- a. stratified squamous, nonkeratinized, for protection.
- b. simple squamous, for rapid diffusion across the epithelium.
- c. simple columnar, for secretion and active absorption.
- d. simple cuboidal, like that of most other tubular structures.
- e. regionally specialized for regionally localized functions.

9.

10. 8

11.

12. 14. The myenteric (Auerbach's) plexus is located:

- a. just beneath the epithelium of the mucosa, in lamina propria.
- b. within lamina propria, adjacent to the muscularis mucosae.
- c. in submucosa.
- d. between the layers of smooth muscle in the muscularis.
- e. in serosa and/or adventitia.

13. 15. The myenteric (Auerbach's) plexus does *not* contain:

- a. postganglionic axons from sympathetic ganglia.
- b. postganglionic axons from parasympathetic ganglia.
- c. nerve cell bodies in sympathetic ganglia.
- d. nerve cell bodies in parasympathetic ganglia.
- e. preganglionic parasympathetic axons.

14. 16. For an organ lying within a body cavity, the tissue which forms the deepest layer (i.e., farthest from the inner core or lumen of the organ) is called the:

- a. mucosa.
- b. serosa.
- c. stroma.
- d. parenchyma.
- e. adventitia.

29. In the digestive system, lymph nodules occur:

- a. concentrated in the tonsils.
- b. concentrated in the appendix.
- c. scattered in lamina propria throughout the GI tract.
- d. concentrated in Peyer's patches.
- e. all of the above.

30. The papillae covering most of the top surface of the human tongue (the tips of these papillae are whitish in life) are called:

- a. tonsils.
- b. fungiform papillae.
- c. filiform papillae.
- d. circumvallate papillae.
- e. dermal papillae.

31. The whitish tips of filiform papillae consist of:

- a. keratinized epithelial tissue.

- b. collagen.
- c. mucous.
- d. taste buds.
- e. goblet cells.

32. Taste buds occur most commonly:

- a. on the tips of filiform papillae.
- b. on the sides of fungiform and circumvallate papillae.
- c. in the crypts of lingual tonsils.
- d. only on the tip of the tongue.
- e. anywhere in the stratified squamous epithelium of the tongue.

33. Tubular invaginations (crypts, pits or glands) occurring within the mucosa (embedded in lamina propria), are not characteristic of the:

- a. esophagus
- b. stomach
- c. duodenum
- d. ileum and jejunum
- e. colon

34. Submucosal glands occur in the submucosa of the:

- a. esophagus.
- b. stomach.
- c. jejunum.
- d. ileum.
- e. colon.

35. Submucosal glands also occur in the:

- a. duodenum.
- b. ileum.
- c. jejunum.
- d. appendix.
- e. colon.

36. Folds in the mucosa of the stomach are called:

- a. rugae.
- b. plicae.
- c. villi.
- d. pits.
- e. crypts.

37. Permanent folds in the wall of the intestine, containing a core of submucosa, are called:

- a. rugae.
- b. plicae.
- c. villi.
- d. pits.
- e. crypts.

38. Microscopic finger-like projections of the intestinal mucosa are called intestinal:

- a. rugae.

- b. plicae.
- c. villi.
- d. pits.
- e. crypts.

39. Microscopic indentations in the epithelial surface of the stomach are called gastric:

- a. rugae.
- b. plicae.
- c. villi.
- d. pits.
- e. crypts.

40. Simple tubular glands within the mucosa of the small or large intestine are called intestinal:

- a. rugae.
- b. plicae.
- c. villi.
- d. pits.
- e. crypts.

41. Lymphatic vessels within intestinal villi, which transport lipids to the circulatory system, are called:

- a. capillaries.
- b. ductules.
- c. canaliculi.
- d. lacteals.
- e. crypts.

42. Taenia coli are:

- a. discrete bands of longitudinal muscle along the large intestine.
- b. crypts or mucosal glands of the large intestine.
- c. submucosal glands of the large intestine.
- d. Compact regions of lymphoid tissue in the intestine.
- e. sites of mucosal folding into the lumen of the large intestine.

-

43. Choose the best description of the esophagus.

- a. Nonkeratinized stratified squamous epithelium; relatively unspecialized lamina propria; submucosa with scattered mucous glands.
- b. Keratinized stratified squamous epithelium; relatively unspecialized lamina propria; submucosa with scattered serous glands.
- c. Nonkeratinized stratified squamous epithelium; lamina propria with numerous lymph nodules; submucosa with scattered mucous glands.
- d. Keratinized stratified squamous epithelium; lamina propria with numerous lymph nodules; are submucosa with scattered mucous glands.
- e. Keratinized stratified squamous epithelium; relatively unspecialized lamina propria; submucosa with scattered mucous glands.

44. Choose the best description of the duodenum.

- a. Mucosa with crypts and villi; epithelium with scattered goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.
- b. Mucosa with crypts but no villi; epithelium with scattered goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.
- c. Mucosa with both crypts and villi; epithelium with scattered goblet cells; lamina propria with occasional lymph nodules; submucosa with mucous glands.
- d. Mucosa with both crypts and villi; epithelium with scattered goblet cells; lamina propria packed with mucous glands; submucosa with numerous lymph nodules.
- e. Mucosa with crypts but no villi; epithelium with scattered goblet cells; lamina propria with occasional lymph nodules; submucosa with mucous glands.

45. Which of the following epithelial types is most closely associated with active absorption at a mucosal surface, as in small intestine and colon?

- a. simple squamous
- b. stratified squamous
- c. simple columnar
- d. stratified columnar
- e. stratified cuboidal

46. Which of the following epithelial types is most closely associated with active secretion at a mucosal surface, as in the stomach lining?

- a. simple squamous
- b. stratified squamous
- c. simple columnar
- d. stratified columnar
- e. stratified cuboidal

47. Choose the best description of the fundic stomach.

- a. Mucosa without villi, but thick and densely packed with tubular glands containing several distinct cell types; epithelium apparently consisting of a single cell type lining surface and pits; lamina propria with numerous lymph nodules.
- b. Mucosa without villi, but thick and densely packed with tubular glands containing several distinct cell types; epithelium apparently consisting of a single cell type lining surface and pits; lamina propria inconspicuous.
- c. Mucosa with villi, thick and densely packed with tubular glands containing several distinct cell types; epithelium apparently consisting of a single cell type lining surface and pits; lamina propria inconspicuous.
- d. Mucosa with villi, thick and densely packed with tubular glands containing several distinct cell types; epithelium with scattered goblet cells lining surface and pits; lamina propria inconspicuous.
- e. Mucosa with villi, thick and densely packed with tubular glands containing several distinct cell types; epithelium with scattered goblet cells lining surface and pits; lamina propria with numerous lymph nodules.

48. Choose the best description of the jejunum.

- a. Mucosa with both crypts and villi; epithelium with scattered goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.
- b. Mucosa with crypts but no villi; epithelium with scattered goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.
- c. Mucosa with both crypts and villi; epithelium with scattered goblet cells; lamina propria with occasional lymph nodules; submucosa with mucous glands.
- d. Mucosa with both crypts and villi; epithelium uniform, without goblet cells; lamina propria with occasional lymph nodules; submucosa with numerous lymph nodules.

e. Mucosa with crypts but no villi; epithelium uniform, without goblet cells; lamina propria with occasional lymph nodules; submucosa with mucous glands.

49. Choose the best description of the ileum.

a. Mucosa with both crypts and villi; epithelium uniform, without goblet cells; lamina propria with occasional conspicuous clusters of lymph nodules; submucosa unspecialized.

b. Mucosa with crypts but no villi; epithelium with scattered goblet cells; lamina propria with occasional conspicuous clusters of lymph nodules; submucosa unspecialized.

c. Mucosa with both crypts and villi; epithelium with scattered goblet cells; lamina propria with occasional conspicuous clusters of lymph nodules; submucosa with mucous glands.

d. Mucosa with both crypts and villi; epithelium with scattered goblet cells; lamina propria with occasional conspicuous clusters of lymph nodules; submucosa unspecialized.

e. Mucosa with both crypts and villi; epithelium with scattered goblet cells; lamina propria with lymph nodules relatively uncommon; submucosa with mucous glands.

50. Choose the best description of the colon.

a. Mucosa with villi and short tubular invaginations; epithelium with many goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.

b. Mucosa with short tubular invaginations but no villi; epithelium consisting entirely of goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.

c. Mucosa with villi and short tubular invaginations; epithelium consisting entirely of goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.

d. Mucosa thick, with many tubular glands consisting of several distinct secretory cell types; epithelium with many goblet cells; lamina propria inconspicuous; submucosa unspecialized.

e. Mucosa with short tubular invaginations but no villi; epithelium with many goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.

51. Which of the following epithelial types is most closely associated with passive protection, without specialization for either secretion or absorption, as in the esophagus and anal canal?

a. simple squamous

b. stratified squamous

c. simple columnar

d. stratified columnar

e. stratified cuboidal

52. Which of the following regions of the GI tract is not characterized by simple columnar epithelium on the mucosal surface?

a. stomach

b. small intestine

c. appendix

d. large intestine

e. anal canal

53. Choose the best description of the appendix.

a. Mucosa with short tubular invaginations but no villi; epithelium lacking goblet cells; lamina propria packed with lymphoid tissue containing numerous germinal centers.

b. Mucosa lacking both tubular invaginations and villi; epithelium with many goblet cells; lamina propria unspecialized, without lymphoid tissue.

c. Mucosa lacking both tubular invaginations and villi; epithelium lacking goblet cells; lamina propria packed with lymphoid tissue containing numerous germinal centers.

d. Mucosa with short tubular invaginations but no villi; epithelium with many goblet cells; lamina propria unspecialized, without lymphoid tissue.

e. Mucosa with short tubular invaginations but no villi; epithelium with many goblet cells; lamina propria packed with lymphoid tissue containing numerous germinal centers.

54. Choose the best description of the gall bladder. (If you haven't learned gall bladder yet, choose the answer which does not match any region of the GI tract proper.)

a. Mucosa with evaginations resembling villi (actually wrinkles), but no crypts; simple columnar epithelium consisting of a single absorptive cell type, without goblet cells; lamina propria unspecialized; no muscularis mucosa.

b. Mucosa with short tubular invaginations but no villi; epithelium with many goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.

c. Nonkeratinized stratified-squamous epithelium; lamina propria unspecialized; distinct muscularis mucosa; submucosa with scattered mucous glands.

d. Mucosa without villi, but thick and densely packed with tubular glands containing several distinct cell types; epithelium apparently consisting of a single cell type lining surface and pits; lamina propria inconspicuous.

e. Mucosa with short tubular invaginations but no villi; epithelium with many goblet cells; lamina propria packed with lymphoid tissue containing numerous germinal centers.

55. Which of the following regions of the GI tract is characterized by simple columnar epithelium on the mucosal surface which apparently consists of a single cell type, without goblet cells?

a. esophagus

b. stomach

c. small intestine

d. large intestine

e. appendix

56. In which of the following regions of the GI tract does the highest proportion of goblet cells occur in the surface epithelium of the mucosa?

a. stomach

b. duodenum

c. jejunum

d. ileum

e. large intestine

57. Which of the following regions of the GI tract is characterized by short tubular invaginations (crypts or pits) which are not lined by dividing, differentiating cells?

a. esophagus

b. stomach

c. duodenum

d. jejunum

e. ileum

58. Which of the following regions of the GI tract is characterized by numerous short tubular invaginations but no villi?

a. duodenum

b. jejunum

c. ileum

d. colon

e. anal canal

59. Which of the following regions of the GI tract is characterized by stratified squamous epithelium?

- ☐ a. duodenum
- ☐ b. jejunum
- ☐ c. ileum
- ☐ d. colon
- ☐ e. anal canal

60. Which of the following regions of the GI tract is characterized by villi?

- ☐ a. esophagus
- ☐ b. stomach
- ☐ c. small intestine
- ☐ d. large intestine
- ☐ e. appendix

61. Which of the following regions of the GI tract is characterized by permanent folds (plica)?

- ☐ a. esophagus
- ☐ b. stomach
- ☐ c. small intestine
- ☐ d. large intestine
- ☐ e. appendix

62. Which of the following regions of the GI tract is characterized by a thick mucosa densely packed with tubular glands containing a variety of cell types?

- ☐ a. esophagus
- ☐ b. stomach
- ☐ c. small intestine
- ☐ d. large intestine
- ☐ e. appendix

63. In which of the following regions of the GI tract does the mucosa most closely resemble that of the colon, but with more lymphoid tissue?

- ☐ a. stomach
- ☐ b. duodenum
- ☐ c. jejunum
- ☐ d. ileum
- ☐ e. appendix

64. In which region(s) of the GI tract are there tubular mucous glands in the mucosa?

- ☐ a. upper esophagus
- ☐ b. cardiac and pyloric stomach
- ☐ c. duodenum
- ☐ d. ileum
- ☐ e. appendix

65. In which region of the GI tract is the longitudinal muscle of the muscularis arranged into distinct longitudinal bundles (taenia)?

- ☐ a. stomach
- ☐ b. duodenum
- ☐ c. jejunum

d. ileum

e. colon

68. The parotid gland is a:

a. serous gland, secreting a watery solution of enzymes and immunoglobulins but not mucus.

b. mucous gland, secreting mucus but not digestive enzymes.

c. mixed gland, producing both serous and mucous secretions.

69. The submandibular and sublingual glands are:

a. serous glands, secreting a watery solution of enzymes and immunoglobulins but not mucus.

b. mucous glands, secreting mucus but not digestive enzymes.

c. mixed glands, producing both serous and mucous secretions.

70. Minor salivary glands are mixed, secreting mainly mucus but also some serous product. Minor salivary glands are found:

a. in the cheeks (buccal glands).

b. in the lips (labial glands).

c. in the soft palate (palatine glands).

d. in the tongue (lingual glands).

e. all of the above.

71. Tissue from which of the following is often poorly preserved in autopsy specimens, because it self-destructs soon after death due to post-mortem release of its stored secretory product of proteolytic enzymes?

a. submaxillary and sublingual glands

b. esophageal glands

c. duodenal (Brunner's) glands

d. parotid glands.

e. pancreas

72. The parotid gland and the pancreas are both compound acinar glands, served by a branching duct system. Morphological differences between parotid and pancreas include which of the following?

a. Islets of endocrine tissue occur in the pancreas but not the parotid.

b. Adipocytes are scattered commonly through the stroma of the parotid but not that of the pancreas.

c. Centroacinar cells (initial duct cells appearing within acini) occur in most pancreatic acini but not those of parotid.

d. all of the above.

73. Esophageal glands are located:

a. in the mucosa.

b. in the submucosa.

c. in the serosa.

d. in the adventitia.

e. deep to the wall of the esophagus proper.

74. Gastric glands are located:

a. in the mucosa of the stomach.

- b. in the submucosa of the stomach.
- c. in the serosa of the stomach.
- d. deep to the stomach wall proper.
- e. more than one of the above.

75. Duodenal (Brunner's) glands are located:

- a. in the mucosa of the duodenum.
- b. in the submucosa of the duodenum.
- c. in the adventitia of the posterior duodenal wall.
- d. deep to the wall of the duodenum.
- e. more than one of the above.

76. The pancreas is located:

- a. in the mucosa of the small intestine.
- b. in the submucosa of the small intestine.
- c. in the serosa of the small intestine.
- d. deep to the wall of the small intestine proper.

77. Intestinal crypts of Lieberkühn are located:

- a. in the mucosa of the small intestine.
- b. in the submucosa of the small intestine.
- c. in the serosa of the small intestine.
- d. deep to the wall of the small intestine proper.
- e. more than one of the above.

78. All of the following are primarily mucous-secreting glands EXCEPT:

- a. esophageal glands.
- b. cardiac glands.
- c. fundic glands.
- d. pyloric glands.
- e. duodenal (Brunner's) glands.

79. Which of the following lacks clearly differentiated ducts (i.e., consists almost entirely of secretory cells)?

- a. minor salivary glands
- b. parotid gland
- c. esophageal glands
- d. gastric glands
- e. pancreas

80. Which of the following glands is primarily a protected site for mitotic activity rather than a tissue specialized for secretion?

- a. parotid salivary glands.
- b. esophageal glands.
- c. gastric (fundic) glands.
- d. duodenal (Brunner's) glands.
- e. intestinal crypts (of Lieberkühn).

81. Which of the dimensions is about 1-2 millimeters? (Think about actual sizes as you can see them in gross lab and with your unaided eye on microscope slides, and also the sizes of familiar cells like the RBC. Remember,

having a reasonable intuition about size can help you notice many significant departures from normal or expected structure.)

- a. The thickness of the stomach wall (all layers).
- b. The thickness of the gastric mucosa.
- c. The height of the stomach's surface epithelial cells.
- d. The diameter of a parietal cell.
- e. The length of microvilli.

84. The principal secretory product for cells of Brunner's (duodenal) glands is:

- a. acid mucus.
- b. alkaline mucus.
- c. proteolytic and other digestive enzymes.
- d. lysosomal enzymes.
- e. any one of several duodenal hormones.

-

85. Which of the following is characteristic of mucus-secreting cells?

- a. polarized cytoplasm, with distinct apical and basal regions
- b. rough endoplasmic reticulum in basal cytoplasm
- c. supranuclear Golgi apparatus
- d. large, mucus-containing vesicles occupying apical cytoplasm
- e. all of the above

86. Which the following stains is used to demonstrate mucus (as well as basement membranes and glycogen)?

- a. hematoxylin
- b. eosin
- c. periodic acid Schiff (PAS)
- d. Golgi silver
- e. Mucus can only be stained immunohistochemically.

87. In H&E stained tissue sections, the mucus-containing secretory vesicles in mucous cells typically appear:

- a. red or pink.
- b. blue or purple.
- c. yellow or brown.
- d. black.
- e. clear or only weakly stained.

88. Which of the following is characteristic of serous (enzyme-secreting) cells?

- a. polarized cytoplasm, with distinct apical and basal regions
- b. extensive rough endoplasmic reticulum in basal cytoplasm
- c. supranuclear Golgi apparatus
- d. zymogen vesicles occupying apical cytoplasm
- e. all of the above

90. Contractile epithelial cells which, under autonomic neural control, help expel product from glandular acini and ducts are called:

- a. myoepithelial cells.
- b. mucous cells.
- c. interlobular duct cells.
- d. intralobular duct cells.
- e. enteroendocrine cells.

92. Cells which secrete HCl into the stomach are called:

- a. goblet cells.
- b. chief (zymogenic) cells.
- c. parietal (oxyntic) cells.
- d. enterocytes.
- e. Paneth cells.

93. Cells which secrete pepsinogen into the stomach are called:

- a. goblet cells.
- b. chief (zymogenic) cells.
- c. parietal (oxyntic) cells.
- d. Kupffer cells.
- e. Paneth cells.

94. Enteroendocrine cells, an inconspicuous but important epithelial cell type, occur:

- a. only in the stomach (especially its pyloric region).
- b. only in the small intestine.
- c. only in the colon.
- d. only in the gall bladder.
- e. mainly in the stomach (especially its pyloric region) and in the small intestine, but also elsewhere along the GI tract (and also along respiratory passages).

100. The principal function for intestinal goblet cells is:

- a. secreting digestive enzyme.
- b. secreting one of several various hormones.
- c. secreting mucus.
- d. absorbing nutrients.
- e. mitotic activity for continuing replacement of surface epithelium.

101. The principal function for most surface epithelial cells in the small intestine (enterocytes) is:

- a. secreting digestive enzyme.
- b. secreting one of several various hormones.
- c. secreting mucus.
- d. absorbing nutrients.
- e. mitotic activity for continuing replacement of surface epithelium.

102. The principal function for intestinal enteroendocrine cells is:

- a. secreting digestive enzyme.
- b. secreting one of several various hormones.
- c. secreting mucus.
- d. absorbing nutrients.
- e. mitotic activity for continuing replacement of surface epithelium.

103. The principal function for cells lining the sides of intestinal crypts of Lieberkühn is:

- a. secreting digestive enzyme.
- b. secreting one of several various hormones.
- c. secreting mucus.
- d. absorbing nutrients.
- e. supplying replacement cells for epithelium of villi.

104. The principal secretory product for the Paneth cells clustered at the ends of intestinal crypts is:

- a. mucus.
- b. pepsinogen.
- c. trypsinogen.
- d. any one of several intestinal hormones.
- e. lysosomal enzymes.

105. The principal secretory product for the epithelial cells forming pancreatic islets (of Langerhans) is:

- a. mucus.
- b. pepsinogen.
- c. trypsinogen.
- d. any one of several islet hormones.
- e. lysosomal enzymes.

106. The principal secretory product for the pancreatic acinar cells is:

- a. acid mucus.
- b. alkaline mucus.
- c. proteolytic and other digestive enzymes.
- d. any one of several acinar hormones.
- e. lysosomal enzymes.

107. The principal secretory product for cells of Brunner's (duodenal) glands is:

- a. acid mucus.
- b. alkaline mucus.
- c. proteolytic and other digestive enzymes.
- d. any one of several acinar hormones.
- e. lysosomal enzymes.

108. Undifferentiated, mitotically active epithelial cells which can replace or regenerate the epithelial lining of the small intestine are located:

- a. at the tips of villi.
- b. along the entire villous surface.
- c. in the lamina propria.
- d. along the wall of intestinal crypts.
- e. at the ends (bottoms) of intestinal crypts.

109. Undifferentiated, mitotically active epithelial cells which can replace or regenerate the epithelial lining of the stomach are located:

- a. among mucous cells of the gastric surface.
- b. along the sides of gastric pits.
- c. in the lamina propria.
- d. in the neck region of gastric glands.
- e. at the deep ends (bottoms) of gastric glands.

122. The structural unit of liver tissue, centered around a central vein with branches of the hepatic artery, portal vein and bile duct at the corners, is called a:

- a. liver acinus.
- b. portal triad, tract or canal.
- c. classical or hepatic lobule.
- d. hepatic cord.
- e. biliary tract.

123. The hepatic lobule can also be defined as a region of hepatic tissue that is served by a single:

- a. portal vein.
- b. central vein.
- c. bile duct.
- d. hepatic artery.
- e. sinusoid.

124. The tissue region located at the corners of a classical liver lobule, containing branches of portal vein, hepatic artery and bile duct all enveloped in connective tissue, is called a:

- a. liver acinus.
- b. portal triad, tract or canal.
- c. classical or hepatic lobule.
- d. hepatic cord.
- e. biliary tract.

125. A functional unit of liver tissue served by a single terminal branch of the hepatic artery, including all the hepatic cords leading toward central veins on both sides of this arteriole, is called a:

- a. liver acinus.
- b. portal triad, tract or canal.
- c. classical or hepatic lobule.
- d. sinusoid.
- e. biliary tract.

126. Cirrhosis, or liver scarring in which functional hepatic tissue is replaced by connective tissue, can be recognized by:

- a. a greater than normal amount of connective tissue in the liver.
- b. continuous strands of connective tissue extending between portal areas.
- c. multiple bile ducts embedded in large portal areas (where the intervening hepatic parenchyma has disappeared).
- d. by using a trichrome stain, which highlights collagen.
- e. by all of the above.

127. The vessel which brings venous blood from the intestine and spleen into the liver, with branches in each portal area, is the:

- a. hepatic artery.
- b. portal vein.
- c. central vein.
- d. sinusoid.
- e. bile duct.

128. The vessel which brings oxygenated, arterial blood from the heart into the liver, with branches in each portal area, is the:

- a. [hepatic artery.](#)
- b. [portal vein.](#)
- c. [central vein.](#)
- d. [sinusoid.](#)
- e. [bile duct.](#)

129. The vessel which drains bile from the liver, with branches in each portal area, is the:

- a. [hepatic artery.](#)
- b. [portal vein.](#)
- c. [central vein.](#)
- d. [sinusoid.](#)
- e. [bile duct.](#)

130. The vessel which drains venous blood from the center of each classical lobule is the:

- a. [hepatic artery.](#)
- b. [portal vein.](#)
- c. [central vein.](#)
- d. [sinusoid.](#)
- e. [bile duct.](#)

131. A hepatic cord is:

- a. [another name for a branch of the portal vein.](#)
- b. [the tough connective tissue surrounding a portal triad.](#)
- c. [a row of adjacent hepatocytes within a liver lobule.](#)
- d. [a branch of the bile duct.](#)
- e. [a nerve within the liver.](#)

132. A vascular passageway between adjacent hepatic cords is called a:

- a. [hepatic artery.](#)
- b. [portal vein.](#)
- c. [central vein.](#)
- d. [sinusoid.](#)
- e. [bile duct.](#)

133. The term liver parenchyma refers to:

- a. [hepatocytes.](#)
- b. [portal veins and hepatic arteries.](#)
- c. [bile ducts.](#)
- d. [stored glycogen.](#)
- e. [connective tissue in the liver.](#)

134. Hepatic sinusoids are lined by:

- a. [nothing; they are free spaces between hepatic cords.](#)
- b. [stratified squamous epithelium.](#)
- c. [endothelium, like any other vascular channel.](#)
- d. [simple cuboidal epithelium.](#)
- e. [simple columnar epithelium.](#)

135. The endothelium lining hepatic sinusoids is:

- a. continuous.
- b. fenestrated.
- c. open-ended.
- d. impermeable to large molecules.
- e. nonexistent.

136. What fluid flows within hepatic sinusoids?

- a. bile
- b. lymph
- c. venous blood from intestine and spleen
- d. arterial blood from hepatic artery
- e. mixed arterial and venous blood

137. Between liver sinusoidal endothelium and the basal surfaces of hepatocytes is the:

- a. bile canaliculus.
- b. space of Disse.
- c. sinusoidal lumen.
- d. portal canal.
- e. central vein.

138. The basal surface of hepatocytes is bathed by blood plasma as it flows freely through:

- a. bile canaliculi.
- b. portal canals.
- c. the space of Disse.
- d. the central veins.
- e. the portal veins.

139. Blood cells (both RBCs and leukocytes) are retained by the sinusoidal endothelium and do not normally enter the:

- a. hepatic artery.
- b. lumen of hepatic sinusoids.
- c. space of Disse.
- d. central veins.
- e. portal veins.

140. A tiny passage between adjacent hepatocytes, lying entirely within a hepatic cord, is called a:

- a. bile canaliculus.
- b. space of Disse.
- c. sinusoidal lumen.
- d. portal canal.
- e. central vein.

141. Kupffer cells are most similar to:

- a. lymphocytes.
- b. hepatocytes.
- c. neutrophils.
- d. macrophages.
- e. bile canaliculi.

142. Kupffer cells are found most commonly within the:

- a. gall bladder.
- b. portal canals.
- c. hepatic sinusoids.
- d. central vein.
- e. bile ducts.

143. Hepatocyte function includes all of the following EXCEPT:

- a. drug detoxification.
- b. protein synthesis.
- c. gluconeogenesis.
- d. urea formation by deamination of amino acids.
- e. breakdown of hemoglobin.

144. Macrophages in the spleen and Kupffer cells in the liver perform which of the following functions?

- a. drug detoxification
- b. antibody secretion
- c. mass storage of glycogen
- d. exocrine secretion
- e. breakdown of hemoglobin

145. Hepatocytes are:

- a. lymphoid tissue.
- b. epithelial tissue.
- c. muscle tissue.
- d. adipose tissue.
- e. nervous tissue.

146. The shape of hepatocytes is best described as:

- a. cuboidal.
- b. columnar.
- c. squamous.
- d. fusiform.
- e. stellate, with many branches.

147. Hepatocytes are characterized by:

- a. round nuclei (occasionally paired).
- b. plentiful rough and smooth endoplasmic reticulum.
- c. many mitochondria.
- d. numerous lysosomes.
- e. all of the above.

149. Endocrine secretory products of hepatocytes include all of the following EXCEPT:

- a. glucose (from glycogen).
- b. serum albumin.
- c. prothrombin.
- d. conjugated bilirubin and bile acids.
- e. fibrinogen.

150. Exocrine secretory products of hepatocytes include which of the following?

- a. glucose (from glycogen)
- b. serum albumin
- c. prothrombin
- d. conjugated bilirubin and bile acids
- e. fibrinogen

151. Fat-storing stellate cells (Ito cells) in the space of Disse are noted for storing:

- a. glycogen.
- b. glucose.
- c. albumin.
- d. vitamin A.
- e. all of the above.

152. Hepatocytes synthesize:

- a. glycogen.
- b. cholesterol.
- c. albumin.
- d. glucose.
- e. all of the above.

153. The gall bladder is best understood as a specialized region of:

- a. a lymph node.
- b. a duct.
- c. adipose tissue.
- d. hepatocyte parenchyma.
- e. a sympathetic ganglion.

154. Like the small intestine, the gall bladder:

- a. is lined by columnar cells with a brush border of microvilli.
- b. has goblet cells in the mucosal epithelium.
- c. has crypts with Paneth cells at the bottom.
- d. has villi with lacteals in the core.
- e. has a distinct muscularis mucosa, and is further surrounded by two distinct layers of circular and longitudinal muscle.

155. Although the gall bladder lining superficially resembles the mucosa of the small intestine, the gall bladder lacks:

- a. an epithelial lining.
- b. columnar cells.
- c. microvilli.
- d. lamina propria.
- e. a muscularis mucosa, crypts, villi, and goblet cells.

The most numerous cell type in the olfactory epithelium, whose nuclei are positioned at several levels in the lower portion of the epithelium, is the:

- a. olfactory receptor cell.*
- b. goblet cell.*
- c. sustentacular cell.*

d. clara cell.

e. basal cell.

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2. Olfactory receptor cells:

a. have several extremely long cilia for extensive exposure of receptor surface membrand.

b. can be replaced by cell division, when worn out or damaged.

c. are supported by sustentacular cells.

d. are nerve cells with axons extending through the cribriform plate into the olfactory bulb of the brain.

e. all of the above

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3. Nasal mucosa includes:

a. pseudostratified, ciliated columnar epithelium.

b. lamina propria.

c. seromucous glands.

d. venus plexuses.

e. all of the above

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4. Small sero-mucous glands may be found in the mucosa of the:

a. nasal cavity.

b. nasal sinuses.

c. trachea.

d. bronchi.

e. all of the above

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5. In which of the following are goblet cells most frequent?

a. trachea

b. bronchus

c. bronchiole

d. alveolus

e. The frequency of goblet cells in each of the above is about equal.

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6. As respiratory passages branch from trachea to bronchioles, the epithelium gradually changes from:

a. simple squamous to ciliated, pseudostratified columnar.

b. simple columnar to simple squamous.

c. simple cuboidal to simple columnar.

d. stratified squamous to simple squamous.

e. ciliated, pseudostratified columnar to simple cuboidal.

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7. Simple squamous epithelium lines the:

a. nasal cavities.

b. crypts of pharyngeal tonsils.

c. bronchi.

d. trachea.

e. alveoli.

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8. Alveolar type I cells are:

- a. cuboidal cells that secrete surfactant.
- b. squamous cells involved in gas exchange.
- c. ciliated cells that move mucous.
- d. columnar cells that secrete mucous.
- e. amoeboid, phagocytic cells that clean the alveolar surface.

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9. Alveolar type II cells are:

- a. cuboidal cells that secrete surfactant.
- b. squamous cells involved in gas exchange.
- c. ciliated cells that move mucous.
- d. columnar cells that secrete mucous.
- e. amoeboid, phagocytic cells that clean the alveolar surface.

-

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10. Alveolar dust cells (macrophages) are:

- a. cuboidal cells that secrete surfactant.
- b. squamous cells involved in gas exchange.
- c. ciliated cells that move mucous.
- d. columnar cells that secrete mucous.
- e. amoeboid that clean the alveolar surface.

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11. The entire alveolar wall or septum (separating adjacent alveoli) consists of:

- a. surface epithelium.
- b. blood vessels.
- c. connective tissue.
- d. all of the above

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12. Non-ciliated cuboidal cells in bronchiole epithelium that are thought to be involved in absorption or secretion are called:

- a. chief cells.
- b. clara cells.
- c. goblet cells.
- d. basal cells.
- e. Type I cells.

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13. The thickness of interalveolar septa separating adjacent alveoli is typically:

- a. 1 micron or less.
- b. 5 to 10 microns.
- c. 50 to 100 microns.
- d. 500 to 1000 microns.

e. greater than 5 mm.

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14. The lining of the pleural cavity is:

a. endothelium.

b. mesothelium.

c. pseudostratified columnar epithelium.

d. type I pneumocytes.

e. type II pneumocytes.

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15. C-shaped rings that form the framework of the trachea and help keep it open are composed of:

a. fibro-elastic tissue.

b. trachealis muscle.

c. skeletal muscle.

d. hyaline cartilage.

e. bone.

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16. Which of the following is devoid of hyaline cartilage?

a. trachea

b. bronchus

c. larynx

d. bronchiole

e. none of the above

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1. One lobe of the kidney:

a. is separated from other lobes by connective tissue septa.

b. is visible grossly as a rounded bulge on the kidney surface.

c. consists of one medullary pyramid together with the associated cortex.

d. drains tubular fluid into a single interlobular duct.

e. is all of the above.

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2. The region of the kidney containing glomeruli is called the:

a. cortex.

b. medulla.

c. calyx.

d. pelvis.

e. papillae.

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3. The region of the kidney containing proximal convoluted tubules is called the:

a. cortex.

b. medulla.

c. calyx.

d. pelvis.

- e. [papillae.](#)
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- 4. The region of the kidney containing long loops of Henle is called the:
 - a. [cortex.](#)
 - b. [medulla.](#)
 - c. [calyx.](#)
 - d. [pelvis.](#)
 - e. [papillae.](#)-
-
- 5. The region of the kidney containing distal convoluted tubules is called the:
 - a. [cortex.](#)
 - b. [medulla.](#)
 - c. [capsule.](#)
 - d. [pelvis.](#)
 - e. [hilus.](#)-
-
- 6. The region of the kidney containing larger collecting ducts is called the:
 - a. [cortex.](#)
 - b. [medulla.](#)
 - c. [calyx.](#)
 - d. [pelvis.](#)
 - e. [papilla.](#)-
-
- 7. Regions of cortical parenchyma located between medullary pyramids rather than adjacent to the surface of the kidney are called:
 - a. [medullary rays.](#)
 - b. [renal lobules.](#)
 - c. [columns of Bertin.](#)
 - d. [papillae.](#)
 - e. [calyces.](#)-
-
- 8. Regions of cortex with texture similar to that of the medulla, containing straight distal tubules and collecting ducts, are called:
 - a. [medullary rays.](#)
 - b. [renal lobules.](#)
 - c. [columns of Bertin.](#)
 - d. [papillae.](#)
 - e. [calyces.](#)-
-
- 9. A region of cortex surrounding and drained by a single bundle of collecting ducts is called a:
 - a. [medullary ray.](#)
 - b. [renal lobule.](#)
 - c. [column of Bertin.](#)
 - d. [papilla.](#)

e. [calyx.](#)

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10. The end of each medullary pyramid through which collecting ducts open into a calyx is called a:

- a. [medullary ray.](#)
- b. [renal lobule.](#)
- c. [column of Bertin.](#)
- d. [papilla.](#)
- e. [hilus.](#)

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11. The tip of each papilla, perforated by the openings of collecting ducts, is called the:

- a. [medullary ray.](#)
- b. [capsule.](#)
- c. [column of Bertini.](#)
- d. [area cribosa.](#)
- e. [hilus.](#)

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12. The space into which one medullary pyramid drains is called one:

- a. [column.](#)
- b. [papilla.](#)
- c. [calyx.](#)
- d. [pelvis.](#)
- e. [hilus.](#)

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13. The large space lined by transitional epithelium which funnels urine into the ureter is called the:

- a. [capsule.](#)
- b. [papillae.](#)
- c. [calyx.](#)
- d. [pelvis.](#)
- e. [hilus.](#)

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14. The capsule of the kidney consists of:

- a. [transitional epithelium.](#)
- b. [dense fibrous connective tissue.](#)
- c. [loose cellular connective tissue with many lymphocytes.](#)
- d. [adipose connective tissue.](#)
- e. [endothelium.](#)

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15. The region of the kidney where artery, vein and ureter are attached is called the:

- a. [cortex.](#)
- b. [medulla.](#)
- c. [calyx.](#)
- d. [pelvis.](#)

e. [hilus.](#)

-

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16. The renal sinus, a space within the hilus that surrounds the renal pelvis, is filled with:

- a. [extracellular fluid.](#)
- b. [air.](#)
- c. [loose adipose connective tissue.](#)
- d. [urine.](#)
- e. [dense fibrous connective tissue.](#)

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17. A principal function for the glomerulus is:

- a. [filtration of blood from capillaries into urinary space.](#)
- b. [active resorption of nutrients from ultrafiltrate back into the blood.](#)
- c. [Volume reduction tubular fluid, with ions pumped from lumen into intersitium followed by diffusion of water.](#)
- d. [regulated passive resorption of water through the medullary concentration gradient.](#)
- e. [regulation of plasma volume and blood pressure through renin secretion.](#)

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18. A principle function for the proximal convoluted tubule is:

- a. [filtration of blood from capillaries into urinary space.](#)
- b. [active resorption of nutrients from ultrafiltrate back into the blood.](#)
- c. [establishment of medullary concentration gradient.](#)
- d. [regulated passive resorption of water through the medullary concentration gradient.](#)
- e. [regulation of plasma volume and blood pressure through renin secretion.](#)

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19. A principal function for the juxtaglomerular apparatus is:

- a. [filtration of blood from capillaries into the urinary space.](#)
- b. [active resorption of nutrients from ultrafiltrate back into the blood.](#)
- c. [volume reduction of tubular fluid, with ions pumped from lumen into intersitium followed by diffusion of water.](#)
- d. [regulated passive resorption of water through the medullary concentration gradient.](#)
- e. [regulation of plasma volume and blood pressure through renin secretion.](#)

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20. A principal function for the loop of Henle is:

- a. [filtration of blood from capillaries into urinary space.](#)
- b. [active resorption of nutrients from ultrafiltrate back into the blood.](#)
- c. [establishment of the medullary concentration gradient.](#)
- d. [regulated passive resorption of water through the medullary concentration gradient.](#)
- e. [regulation of plasma volume and blood pressure through renin secretion.](#)

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21. A principal function for the distal convoluted tubule is:

- a. [establishment of the medullary concentration gradient.](#)
- b. [active resorption of nutrients from ultrafiltrate back into the blood.](#)
- c. [volume reduction of tubular fluid, with ions pumped from lumen into interstitium followed by diffusion of water.](#)

- d. [regulated passive resorption of water through the medullary concentration gradient.](#)
- e. [regulation of plasma volume and blood pressure through renin secretion.](#)

-

22. Fluid which is transported from the lumen of the proximal and distal convoluted tubules into the cortical interstitium is take up by the:

- a. [collecting ducts.](#)
- b. [pelvis.](#)
- c. [podocytes.](#)
- d. [pertitubular capillaries.](#)
- e. [loops of Henle.](#)

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23. A principle function for the collecting duct is:

- a. [filtration of blood from capillaries into urinary space.](#)
- b. [active resorption of nutrients from ultrafiltrate back into the blood.](#)
- c. [volume reduction tubular fluid, with ions pumped from lumen into interstitium followed by diffusion of water.](#)
- d. [regulated passive resorption of water through the medullary concentration gradient.](#)
- e. [regulation of plasma volume and blood pressure through rennin secretion.](#)

-

24. The region of glomerulus (renal corpuscle) where the capillary tuft is attached is called the:

- a. [mesangium.](#)
- b. [area cribosa.](#)
- c. [vascular pole.](#)
- d. [urinary \(tubular\) pole.](#)
- e. [juxtaglomerular apparatus.](#)

-

25. The region of glomerulus (renal corpuscle) where the urinary space drains into the proximal convoluted tubule is called the:

- a. [mesangium.](#)
- b. [area cribosa.](#)
- c. [vascular pole.](#)
- d. [urinary \(tubular\) pole.](#)
- e. [juxtaglomerular apparatus.](#)

-

26. The lumen of glomerular capillaries is lined by:

- a. [mesangial cells.](#)
- b. [podocytes foot processes.](#)
- c. [juxtaglomerular cells.](#)
- d. [continuous endothelium.](#)
- e. [fenestrated endothelium.](#)

-

27. Interstitial cells located in matrix between glomerular capillaries are called:

- a. [pericytes.](#)
- b. [juxtaglomerular cells.](#)

- c. [podocytes.](#)
- d. [mesangial cells.](#)
- e. [fibroblasts.](#)

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28. Glomerular capillaries are supported on the outside (adjacent to the urinary space) by:

- a. [mesangial cells.](#)
- b. [fibroblasts.](#)
- c. [podocyte foot processes.](#)
- d. [pericytes.](#)
- e. [juxtaglomerular cells.](#)

-

29. Smooth muscle cells of renal arterioles which are modified for endocrine secretion are called:

- a. [mesangial cells.](#)
- b. [fibroblasts.](#)
- c. [podocytes.](#)
- d. [pericytes.](#)
- e. [juxtaglomerular cells.](#)

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30. The outer lining of Bowman's capsule (the parietal epithelium) consists of a single layer of:

- a. [squamous cells.](#)
- b. [cuboidal cells.](#)
- c. [podocytes.](#)
- d. [pericytes.](#)
- e. [mesangial cells.](#)

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31. Filtration slits are located between adjacent:

- a. [layers of the glomerular basement membrane.](#)
- b. [capillary fenestrations.](#)
- c. [podocyte foot processes.](#)
- d. [mesangial cells.](#)
- e. [capillary endothelial cells.](#)

-

32. Podocytes are:

- a. [simple squamous epithelial cells.](#)
- b. [simple cuboidal epithelial cells.](#)
- c. [simple columnar epithelial cells.](#)
- d. [unique epithelial cells with many interdigitating processes.](#)
- e. [not epithelial cells.](#)

-

33. Mesangial cells are:

- a. [simple squamous epithelial cells.](#)
- b. [simple cuboidal epithelial cells.](#)
- c. [simple columnar epithelial cells.](#)
- d. [unique epithelial cells with many interdigitating processes.](#)

- e. [not epithelial cells.](#)

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34. The normal thickness of the glomerular basement membrane is between:

- a. [30 and 35 nanometers.](#)
- b. [0.3 and 0.35 micrometers.](#)
- c. [3 and 3.5 micrometers.](#)
- d. [30 and 35 micrometers.](#)
- e. [300 and 350 micrometers.](#)

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35. A basement membrane is also found beneath which of the following epithelia?

- a. [bowman's capsule.](#)
- b. [proximal convoluted tubule.](#)
- c. [distal convoluted tubule.](#)
- d. [macula densa.](#)
- e. [all of the above.](#)

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36. The component of the glomerular filtration barrier which is permeable to water and small molecules but not larger molecules like proteins is the:

- a. [mesangial cells.](#)
- b. [podocyte foot processes.](#)
- c. [fenestrated endothelium.](#)
- d. [glomerular basement membrane.](#)
- e. [bowman's capsule.](#)

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37. The component of the glomerular filtration barrier which holds back blood cells but allows fluid and large molecules to pass freely is the:

- a. [mesangial cells.](#)
- b. [podocyte foot processes.](#)
- c. [fenestrated endothelium.](#)
- d. [glomerular basement membrane.](#)
- e. [bowman's capsule.](#)

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38. Extraglomerular mesangial cells (laci cells) are located:

- a. [at the vascular pole of the glomerulus.](#)
- b. [at the urinary pole of the glomerulus.](#)
- c. [within the macula densa.](#)
- d. [encircling the proximal tubule.](#)
- e. [between adjacent podocyte foot processes.](#)

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39. Epithelial cells lining the proximal convoluted tubule:

- a. [have an apical brush border of microvilli for active absorption from the lumen.](#)
- b. [have basal infoldings and interdigitations with adjacent cells for increased basal membrane surface area.](#)
- c. [have many mitochondria to provide energy for active transport against a concentration gradient .](#)

- d. [are cuboidal.](#)
- e. [all of the above.](#)

-

40. Epithelial cells lining the distal convoluted tubule:

- a. [have an apical brush border of microvilli for active absorption from the lumen.](#)
- b. [have basal infoldings and interdigitations with adjacent cells for increased basal membrane surface area.](#)
- c. [have many mitochondria to provide energy for active transport against a concentration gradient.](#)
- d. [are cuboidal.](#)
- e. [all of the above *except* the brush border.](#)

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41. The proximal convoluted tubule is:

- a. [several times longer than the distal convoluted tubule.](#)
- b. [about the same length as the distal convoluted tubule.](#)
- c. [less than half the length of the distal convoluted tubule.](#)
- d. [much narrower than the distal convoluted tubule.](#)
- e. [any of the above may be true, depending on location in cortex.](#)

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42. Epithelial cells lining the thin limb of the loop of Henle:

- a. [have an apical brush border of microvilli for active absorption from the lumen.](#)
- b. [contain numerous basally located secretory vesicles.](#)
- c. [contain many ribosomes and rough endoplasmic reticulum for extensive protein synthesis.](#)
- d. [are simple squamous.](#)
- e. [all of the above.](#)

-

43. Epithelial cells lining collecting ducts:

- a. [stain less intensely with H & E than those lining proximal or distal tubules.](#)
- b. [often display distinct intercellular boundaries, unlike those lining proximal or distal tubules.](#)
- c. [regulate their permeability to water in response to antidiuretic hormone concentration.](#)
- d. [are simple cuboidal.](#)
- e. [all of the above.](#)

-

44. The macula densa consists of:

- a. [extraglomerular mesangial cells.](#)
- b. [closely-packed nuclei in the epithelium of the distal convoluted tubule.](#)
- c. [modified smooth muscle cells.](#)
- d. [closely-packed nuclei in the epithelium of the proximal convoluted tubule.](#)
- e. [densely packed collagen fibers in the juxtaglomerular interstitium.](#)

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45. Juxtaglomerular cells secrete:

- a. [urea.](#)
- b. [antidiuretic hormone.](#)
- c. [renin.](#)
- d. [angiotensinogen.](#)

- e. [aldosterone.](#)

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46. Arteries which form arches along the border between the cortex and the medulla of the kidney are called:

- a. [interlobar arteries.](#)
- b. [interlobular arteries.](#)
- c. [arcuate arteries.](#)
- d. [afferent arteries.](#)
- e. [vasa recta.](#)

-

47. Arteries which extend into the renal cortex, usually perpendicular to its surface, are called:

- a. [interlobar arteries.](#)
- b. [interlobular arteries.](#)
- c. [arcuate arteries.](#)
- d. [afferent arteries.](#)
- e. [vasa recta.](#)

-

48. Bundles of parallel vessels which extend into the medulla are called:

- a. [interlobar arteries.](#)
- b. [interlobular arteries.](#)
- c. [arcuate arteries.](#)
- d. [afferent arteries.](#)
- e. [vasa recta.](#)

-

49. Vessels which connect interlobular arteries with glomerular capillaries are called:

- a. [arcuate arteries.](#)
- b. [afferent arterioles.](#)
- c. [efferent arterioles.](#)
- d. [peritubular capillaries.](#)
- e. [vasa recta.](#)

-

50. Vessels which connect glomerular capillaries with peritubular capillaries are called:

- a. [arcuate arteries.](#)
- b. [afferent arterioles.](#)
- c. [efferent arterioles.](#)
- d. [peritubular capillaries.](#)
- e. [vasa recta.](#)

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51. Vessels which surround proximal and distal convoluted tubules, receiving reabsorbed water and nutrients, are called:

- a. [arcuate arteries.](#)
- b. [afferent arterioles.](#)
- c. [efferent arterioles.](#)
- d. [peritubular capillaries.](#)

- e. [vasa recta.](#)

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52. The volume of blood decreases while passing through the:

- a. [interlobular arteries.](#)
- b. [afferent arterioles.](#)
- c. [glomerular capillaries.](#)
- d. [efferent arterioles.](#)
- e. [peritubular capillaries.](#)

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53. The volume of blood increases while passing through the:

- a. [interlobular arteries.](#)
- b. [afferent arterioles.](#)
- c. [glomerular capillaries.](#)
- d. [efferent arterioles.](#)
- e. [peritubular capillaries.](#)

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54. Vessels which form a counter-current exchange, to maintain an osmotic gradient established by long loops of Henle, are called:

- a. [arcuate arteries.](#)
- b. [afferent arterioles.](#)
- c. [efferent arterioles.](#)
- d. [peritubular capillaries.](#)
- e. [vasa recta.](#)

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55. Interstitial fluid in the kidney remains essentially iso-osmotic with blood in each of the following regions EXCEPT:

- a. [cortex, surrounding proximal convoluted tubules.](#)
- b. [cortex, surrounding distal convoluted tubules.](#)
- c. [cortex, within the glomerular mesangium.](#)
- d. [cortex, surrounding Bowman's capsule.](#)
- e. [medulla, surrounding loops of Henle.](#)

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56. Interstitial oxygen concentration is lowest in which of the following regions?

- a. [outer cortex.](#)
- b. [juxtamedullary cortex.](#)
- c. [outer medulla.](#)
- d. [inner medulla, in the papillae.](#)
- e. [interstitial oxygen concentration is similar in all the above areas.](#)

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57. The essential function for the medullary concentration gradient is to permit the:

- a. [passive resorption of nutrients from tubular fluid.](#)
- b. [active resorption of ions from tubular fluid.](#)
- c. [passive resorption of ions from tubular fluid.](#)
- d. [active resorption of water from tubular fluid.](#)

- e. [passive resorption of water from tubular fluid.](#)

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58. The volume of tubular fluid decreases as it passes through which of the following?

- a. [proximal convoluted tubule.](#)
- b. [loop of Henle.](#)
- c. [distal convoluted tubule.](#)
- d. [collecting duct.](#)
- e. [all of the above.](#)

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59. Which segment of the nephron functions to concentrate tubular fluid by passive resorption of water as it as it passes through an osmotic concentration gradient?

- a. [glomerulus.](#)
- b. [proximal convoluted tubule.](#)
- c. [loop of Henle.](#)
- d. [distal convoluted tubule.](#)
- e. [collecting duct.](#)

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60. Which segment of the nephron functions to filter fluid from blood into urinary space?

- a. [glomerulus.](#)
- b. [proximal convoluted tubule.](#)
- c. [loop of Henle.](#)
- d. [distal convoluted tubule.](#)
- e. [collecting duct.](#)

-

61. Which segment of the nephron functions to reduce tubular fluid volume by pumping ions (which are followed by water) into the interstitium?

- a. [glomerular capillaries.](#)
- b. [glomerular mesangium.](#)
- c. [juxtaglomerular apparatus.](#)
- d. [distal convoluted tubule.](#)
- e. [collecting duct.](#)

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62. Which segment of the nephron functions to establish a medullary concentration gradient?

- a. [glomerulus.](#)
- b. [proximal convoluted tubule.](#)
- c. [loop of Henle.](#)
- d. [distal convoluted tubule.](#)
- e. [collecting duct.](#)

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63. Which segment of the nephron functions by active resorption of nutrients from ultrafiltrate back into blood?

- a. [glomerulus.](#)
- b. [proximal convoluted tubule.](#)
- c. [loop of Henle.](#)

- d. [juxtaglomerular apparatus.](#)
- e. [collecting duct.](#)

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64. Which segment of the nephron functions to regulate blood pressure through renin secretion?

- a. [glomerulus.](#)
- b. [proximal convoluted tubule.](#)
- c. [juxtaglomerular apparatus.](#)
- d. [loop of Henle.](#)
- e. [collecting duct.](#)

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65. Which cells respond to aldosterone?

- a. [podocytes.](#)
- b. [epithelial cells of distal tubules.](#)
- c. [juxtaglomerular cells.](#)
- d. [epithelial cells of Bowman's capsule.](#)
- e. [mesangial cells.](#)

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66. Which cells respond to antidiuretic hormone (ADH)?

- a. [podocytes.](#)
- b. [mesangial cells.](#)
- c. [juxtaglomerular cells.](#)
- d. [epithelial cells of Bowman's capsule.](#)
- e. [epithelial cells of collecting duct.](#)

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67. Which cells secrete a product which converts angiotensinogen to angiotensin?

- a. [podocytes.](#)
- b. [epithelial cells of proximal and distal convoluted tubules.](#)
- c. [juxtaglomerular cells.](#)
- d. [epithelial cells of Bowman's capsule.](#)
- e. [epithelial cells of collecting duct.](#)

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68. Which cells secrete angiotensinogen?

- a. [podocytes](#)
- b. [epithelial cells of proximal and distal convoluted tubules.](#)
- c. [juxtaglomerular cells.](#)
- d. [hepatocytes.](#)
- e. [epithelial cells of collecting duct.](#)

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69. The volume of tubular fluid is greatest as it leaves which of the following?

- a. [Bowman's space.](#)
- b. [proximal convoluted tubule.](#)
- c. [loop of Henle.](#)
- d. [distal convoluted tubule.](#)

e. [collecting duct.](#)

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70. Which epithelial cells of the urinary tract are specialized with an apical brush border of microvilli?

- a. [Bowman's capsule epithelium.](#)
- b. [proximal convoluted tubule epithelium.](#)
- c. [epithelium lining the thin segment of the loop of Henle.](#)
- d. [distal convoluted tubule epithelium.](#)
- e. [collecting duct epithelium.](#)

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71. Which epithelial cells of the urinary tract are specialized with basal membrane folds closely associated with many mitochondria?

- a. [Bowman's capsule epithelium.](#)
- b. [proximal and distal convoluted tubule epithelium.](#)
- c. [epithelium lining the thin segment of the loop of Henle.](#)
- d. [collecting duct epithelium.](#)
- e. [transitional epithelium.](#)

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72. Which of the following epithelial cells of the urinary tract are simple squamous, without apparent structural specialization?

- a. [Bowman's capsule epithelium.](#)
- b. [glomerular epithelium \(podocytes\).](#)
- c. [transitional epithelium.](#)
- d. [distal convoluted tubule epithelium.](#)
- e. [collecting duct epithelium.](#)

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73. Which epithelial cells of the urinary tract are specialized with folds of thickened plasma membrane, and including many apical vesicles lined by similar membrane, which may permit rapid expansion of the cells' surface when stretched?

- a. [bowman's capsule epithelium.](#)
- b. [glomerular epithelium \(podocytes\).](#)
- c. [epithelium lining the thin segment of the loop of Henle.](#)
- d. [distal convoluted tubule epithelium.](#)
- e. [transitional epithelium.](#)

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74. Interstitial cells within the renal glomerulus, whose functions are uncertain but may include phagocytosis and maintenance of the glomerular basement membrane, are the:

- a. [podocytes.](#)
- b. [macula densa.](#)
- c. [juxtaglomerular cells.](#)
- d. [mesangial cells.](#)
- e. [transitional epithelium.](#)

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75. Transitional epithelium lines all of the following EXCEPT the:

- a. [renal pelvis.](#)

- b. [ureter.](#)
- c. [bladder.](#)
- d. [proximal urethra.](#)
- e. [Bowman's capsule.](#)

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1.

Which cell is also called a septal cell?

- a. Clara cell
- b. Type I pneumocyte
- c. Type II pneumocyte
- d. Dust cell
- e. Brush cell

Answer: c

The Clara cell is found in the terminal bronchioles. The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte. Brush cells are occasionally, but rarely, seen in the alveolar epithelium.

2.

Which cell is a respiratory macrophage?

- a. Kupffer cells
- b. Histiocyte
- c. Dust cell
- d. Langerhans cell
- e. Microglia

Answer: c

Macrophages are mononuclear phagocytes. Many tissues have resident (fixed) macrophages. Fixed macrophages are given a unique name, depending on the tissue that they are located in. Kupffer cells are the hepatic macrophages. Histiocytes are macrophages seen in connective tissue. Dust cells are alveolar macrophages found in the respiratory tract. Langerhans cells are macrophages seen in the skin. Microglia are the central nervous system macrophages.

3.

Which cell is a type of neuron?

- a. Basal cells
- b. Brush cells
- c. Olfactory cells
- d. Sustentacular cells
- e. All of the above

Answer: c

Olfactory mucosa lines the roof and portions of the walls of the nasal cavity. It contains several cell types: basal cells, brush cells, olfactory cells and sustentacular cells.

Basal cells are located in the basal lamina. Brush cells are involved with general sensation of the olfactory mucosa. Olfactory cells are bipolar neurons that are the receptors for smell. Sustentacular cells are supporting cells. Sustentacular cells are most numerous cell type in the olfactory epithelium.

4.

What cell types are found in the respiratory mucosa?

- a. Ciliated cells*
- b. Goblet cells*
- c. Basal cells*
- d. Brush cells*
- e. All of the above*

Answer: e

In man, respiratory mucosa is composed of ciliated pseudostratified columnar epithelium with goblet cells. Respiratory mucosa contains several cell types: ciliated cells, goblet cells, basal cells, and brush cells. Respiratory mucosa is located lining the respiratory segment of the nasal cavity. It lines the conchae and the paranasal sinuses.

5.

What type of tissue makes up the epiglottis?

- a. Compact bone*
- b. Spongy bone*
- c. Hyaline cartilage*
- d. Fibrocartilage*
- e. Elastic cartilage*

Answer: e

The epiglottis is part of the larynx. It is composed of elastic cartilage.

6.

What is the smallest airspace of the respiratory tree?

- a. Alveolar duct*
- b. Alveoli*
- c. Alveolar sac*
- d. Respiratory bronchiole*
- e. Terminal bronchiole*

Answer: b

The respiratory tract is made of branching structures, much like the branches of a tree. The trachea branches into two bronchi which branch into smaller bronchi. The bronchi ultimately branch into smaller bronchioles. Bronchioles are distinguished from bronchi in that they do not have cartilage and submucosal glands. The terminal bronchioles are the last part of the airway in which gas exchange does not occur. Terminal bronchioles lead to the respiratory bronchioles. The respiratory bronchioles are the first section of the respiratory tree that gas exchange can occur. The alveoli duct is

analogous to a thoroughfare with many cul-du-sacs branching off of it. At the end of the alveoli duct is an alveoli sac. An alveoli sac is a cluster of alveoli, much like a cluster of grapes. Alveoli are individual sacs where gas exchange occurs.

7.

What type of epithelium lines the trachea?

- a. Simple squamous epithelium*
- b. Simple cuboidal epithelium*
- c. Simple columnar epithelium*
- d. Stratified squamous epithelium*
- e. Pseudostratified epithelium*

Answer: e

The trachea is lined by pseudostratified squamous epithelium. Epithelium lines body cavities and surfaces.

Pseudostratified squamous epithelium is "pseudostratified" because it is only one cell layer thick, yet it appears to be stratified. In reality, every cell touches the basement membrane.

Note from Sarah Bellham: The prefix "pseudo" is of Greek origin and it means false or counterfeit. For example: pseudonym, pseudo-science or pseudostratified.

8.

Which structure is part of the conducting portion of the airway?

- a. Bronchi*
- b. Alveolar ducts*
- c. Alveoli*
- d. Alveolar sacs*
- e. Respiratory bronchioles*

Answer: a

The conducting portion of the airway is where air is moved, warmed and moistened. The nasal cavities, pharynx, larynx, trachea and bronchi are all part of the conducting portion of the airway.

The respiratory portion of the airway is where gas exchange occurs. The respiratory bronchioles, alveolar ducts, alveolar sacs and alveoli are all part of the respiratory portion.

9.

What type of epithelium lines the vestibule?

- a. Simple squamous epithelium*
- b. Simple columnar epithelium*
- c. Stratified squamous epithelium*
- d. Ciliated pseudostratified epithelium with goblet cells*
- e. Transitional epithelium*

Answer: c

The vestibule is lined by stratified squamous epithelium.

10.

Which cartilage of the larynx is made of hyaline cartilage?

- a. Thyroid cartilage*
- b. Cricoid cartilage*
- c. Arytenoid cartilage*
- d. Corniculate cartilage*
- e. All of the above*

Answer: e

The larynx is composed of several cartilages. The thyroid cartilage, cricoid cartilage, arytenoid cartilages, corniculate cartilages and cuneiform cartilages are all composed of hyaline cartilage. The epiglottis is elastic cartilage. There is no fibrocartilage in the larynx.

1.

What part of the respiratory tree is the functional unit where gas exchange occurs?

- a. Alveolar duct*
- b. Alveoli*
- c. Alveolar sac*
- d. Respiratory bronchiole*
- e. Terminal bronchiole*

Answer: b

The respiratory tract is made of branching structures, much like the branches of a tree. The trachea branches into two bronchi which branch into smaller bronchi. The bronchi ultimately branch into smaller bronchioles. Bronchioles are distinguished from bronchi in that they do not have cartilage and submucosal glands. The terminal bronchioles are the last part of the airway in which gas exchange does not occur. Terminal bronchioles lead to the respiratory bronchioles. The respiratory bronchioles are the first section of the respiratory tree that gas exchange can occur. The alveoli duct is analogous to a thoroughfare with many cul-du-sacs branching off of it. At the end of the alveoli duct is an alveoli sac. An alveoli sac is a cluster of alveoli, much like a cluster of grapes. Alveoli are individual sacs where gas exchange occurs.

2.

Which cell type is involved in general sensation of the olfactory mucosa?

- a. Basal cells*
- b. Brush cells*
- c. Olfactory cells*
- d. Sustentacular cells*
- e. All of the above*

Answer: b

Olfactory mucosa lines the roof and portions of the walls of the nasal cavity. It contains several cell types: basal cells, brush cells, olfactory cells and sustentacular cells.

Basal cells are located in the basal lamina. Brush cells are involved with general sensation of the olfactory mucosa. Olfactory cells are bipolar neurons that are the receptors for smell. Sustentacular cells are supporting cells. Sustentacular cells are most numerous cell type in the olfactory epithelium.

3.

What type of cells are found in the olfactory mucosa?

- a. Basal cells**
- b. Brush cells**
- c. Olfactory cells**
- d. Sustentacular cells**
- e. All of the above**

Answer: e

Olfactory mucosa lines the roof and portions of the walls of the nasal cavity. It contains several cell types: basal cells, brush cells, olfactory cells and sustentacular cells.

Basal cells are located in the basal lamina. Brush cells are involved with general sensation of the olfactory mucosa. Olfactory cells are bipolar neurons that are the receptors for smell. Sustentacular cells are supporting cells. Sustentacular cells are most numerous cell type in the olfactory epithelium.

4.

What type of tissue makes up the rings of the trachea?

- a. Compact bone**
- b. Spongy bone**
- c. Hyaline cartilage**
- d. Fibrocartilage**
- e. Elastic cartilage**

Answer: c

The rings of the trachea are composed of hyaline cartilage.

5.

Which cell is a squamous pulmonary epithelial cell?

- a. Clara cell**
- b. Type I pneumocyte**
- c. Type II pneumocyte**
- d. Dust cell**
- e. Brush cell**

Answer: b

The Clara cell is found in the terminal bronchioles. The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte. Brush cells are occasionally, but rarely, seen in the alveolar epithelium.

6.

Which cell is also called an alveolar phagocyte?

- a. Clara cell**
- b. Type I pneumocyte**
- c. Type II pneumocyte**
- d. Dust cell**
- e. Brush cell**

Answer: d

The Clara cell is found in the terminal bronchioles. The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte. Brush cells are occasionally, but rarely, seen in the alveolar epithelium.

7.

Which structure is NOT part of the conducting portion of the airway?

- a. Bronchi**
- b. Larynx**
- c. Trachea**
- d. Larynx**
- e. Respiratory bronchioles**

Answer: e

The conducting portion of the airway is where air is moved, warmed and moistened. The nasal cavities, pharynx, larynx, trachea and bronchi are all part of the conducting portion of the airway.

The respiratory portion of the airway is where gas exchange occurs. The respiratory bronchioles, alveolar ducts, alveolar sacs and alveoli are all part of the respiratory portion.

8.

What type of tissue lines the pharynx?

- a. Simple squamous epithelium**
- b. Simple columnar epithelium**
- c. Stratified squamous epithelium**
- d. Ciliated pseudostratified epithelium with goblet cells**
- e. Both c and d**

Answer: e

The pharynx is lined by both stratified squamous epithelium and ciliated pseudostratified epithelium with goblet cells. Different regions are lined by a different type of epithelium. Regions of the pharynx that are likely to be roughened up by food are lined by stratified squamous epithelium. Other regions of the pharynx are lined by ciliated pseudostratified epithelium with goblet cells.

9.

What is an olfactory cell?

- a. Unipolar neuron**
- b. Bipolar neuron**
- c. Multipolar neuron**
- d. Supporting cell**
- e. None of the above**

Answer: b

An olfactory cell is a bipolar neuron.

10.

Which type of cartilage is found in the larynx?

- a. Hyaline cartilage**
- b. Elastic cartilage**
- c. Fibrocartilage**
- d. Both a and b**
- e. All of the above**

Answer: d

The larynx is composed of several cartilages. The thyroid cartilage, cricoid cartilage, arytenoid cartilages, corniculate cartilages and cuneiform cartilages are all composed of hyaline cartilage. The epiglottis is elastic cartilage. There is no fibrocartilage in the larynx.

1.

What part of the respiratory tree is the functional unit where gas exchange occurs?

- a. Alveolar duct**
- b. Alveoli**
- c. Alveolar sac**
- d. Respiratory bronchiole**
- e. Terminal bronchiole**

Answer: b

The respiratory tract is made of branching structures, much like the branches of a tree. The trachea branches into two bronchi which branch into smaller bronchi. The bronchi ultimately branch into smaller bronchioles. Bronchioles are distinguished from bronchi in that they do not have cartilage and submucosal glands. The terminal bronchioles are the last part of the airway in which gas exchange does not occur. Terminal bronchioles lead to the respiratory bronchioles. The respiratory bronchioles are the first section of the respiratory tree that gas exchange can occur. The alveoli duct is analogous to a thoroughfare with many cul-du-sacs branching off of it. At the end of the alveoli duct is an alveoli sac. An alveoli sac is a cluster of alveoli, much like a cluster of grapes. Alveoli are individual sacs where gas

exchange occurs.

2.

Which cell type is involved in general sensation of the olfactory mucosa?

- a. Basal cells**
- b. Brush cells**
- c. Olfactory cells**
- d. Sustentacular cells**
- e. All of the above**

Answer: b

Olfactory mucosa lines the roof and portions of the walls of the nasal cavity. It contains several cell types: basal cells, brush cells, olfactory cells and sustentacular cells.

Basal cells are located in the basal lamina. Brush cells are involved with general sensation of the olfactory mucosa. Olfactory cells are bipolar neurons that are the receptors for smell. Sustentacular cells are supporting cells. Sustentacular cells are most numerous cell type in the olfactory epithelium.

3.

What type of cells are found in the olfactory mucosa?

- a. Basal cells**
- b. Brush cells**
- c. Olfactory cells**
- d. Sustentacular cells**
- e. All of the above**

Answer: e

Olfactory mucosa lines the roof and portions of the walls of the nasal cavity. It contains several cell types: basal cells, brush cells, olfactory cells and sustentacular cells.

Basal cells are located in the basal lamina. Brush cells are involved with general sensation of the olfactory mucosa. Olfactory cells are bipolar neurons that are the receptors for smell. Sustentacular cells are supporting cells. Sustentacular cells are most numerous cell type in the olfactory epithelium.

4.

What type of tissue makes up the rings of the trachea?

- a. Compact bone**
- b. Spongy bone**
- c. Hyaline cartilage**
- d. Fibrocartilage**

e. Elastic cartilage

Answer: c

The rings of the trachea are composed of hyaline cartilage.

5.

Which cell is a squamous pulmonary epithelial cell?

- a. Clara cell**
- b. Type I pneumocyte**
- c. Type II pneumocyte**
- d. Dust cell**
- e. Brush cell**

Answer: b

The Clara cell is found in the terminal bronchioles. The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte. Brush cells are occasionally, but rarely, seen in the alveolar epithelium.

6.

Which cell is also called an alveolar phagocyte?

- a. Clara cell**
- b. Type I pneumocyte**
- c. Type II pneumocyte**
- d. Dust cell**
- e. Brush cell**

Answer: d

The Clara cell is found in the terminal bronchioles. The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte. Brush cells are occasionally, but rarely, seen in the alveolar epithelium.

7.

Which structure is NOT part of the conducting portion of the airway?

- a. Bronchi**
- b. Larynx**
- c. Trachea**
- d. Larynx**
- e. Respiratory bronchioles**

Answer: e

The conducting portion of the airway is where air is moved, warmed and moistened. The nasal cavities, pharynx, larynx, trachea and bronchi are all part of the conducting portion of the airway.

The respiratory portion of the airway is where gas exchange occurs. The respiratory bronchioles, alveolar ducts, alveolar sacs and alveoli are all part of the respiratory portion.

8.

What type of tissue lines the pharynx?

- a. Simple squamous epithelium***
- b. Simple columnar epithelium***
- c. Stratified squamous epithelium***
- d. Ciliated pseudostratified epithelium with goblet cells***
- e. Both c and d***

Answer: e

The pharynx is lined by both stratified squamous epithelium and ciliated pseudostratified epithelium with goblet cells. Different regions are lined by a different type of epithelium. Regions of the pharynx that are likely to be roughened up by food are lined by stratified squamous epithelium. Other regions of the pharynx are lined by ciliated pseudostratified epithelium with goblet cells.

9.

What is an olfactory cell?

- a. Unipolar neuron***
- b. Bipolar neuron***
- c. Multipolar neuron***
- d. Supporting cell***
- e. None of the above***

Answer: b

An olfactory cell is a bipolar neuron.

10.

Which type of cartilage is found in the larynx?

- a. Hyaline cartilage***
- b. Elastic cartilage***
- c. Fibrocartilage***
- d. Both a and b***
- e. All of the above***

Answer: d

The larynx is composed of several cartilages. The thyroid cartilage, cricoid cartilage, arytenoid cartilages, corniculate cartilages and cuneiform cartilages are all composed of hyaline cartilage. The

epiglottis is elastic cartilage. There is no fibrocartilage in the larynx.

Instructions: For each histology question, pick the one best answer. This histology test bank is also useful for the histology questions on the USMLE (USMLE step 1).

1.

What structure is similar to a "bunch of grapes?"

- a. Alveolar duct***
- b. Alveoli***
- c. Alveolar sac***
- d. Respiratory bronchiole***
- e. Terminal bronchiole***

Answer: c

The respiratory tract is made of branching structures, much like the branches of a tree. The trachea branches into two bronchi which branch into smaller bronchi. The bronchi ultimately branch into smaller bronchioles. Bronchioles are distinguished from bronchi in that they do not have cartilage and submucosal glands. The terminal bronchioles are the last part of the airway in which gas exchange does not occur. Terminal bronchioles lead to the respiratory bronchioles. The respiratory bronchioles are the first section of the respiratory tree that gas exchange can occur. The alveoli duct is analogous to a thoroughfare with many cul-du-sacs branching off of it. At the end of the alveoli duct is an alveoli sac. An alveoli sac is a cluster of alveoli, much like a cluster of grapes. Alveoli are individual sacs where gas exchange occurs.

2.

Which cell is found in large numbers in the terminal bronchioles?

- a. Clara cell***
- b. Type I pneumocyte***
- c. Type II pneumocyte***
- d. Dust cell***
- e. Brush cell***

Answer: a

The Clara cell is found in the terminal bronchioles. The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte. Brush cells are occasionally, but rarely, seen in the alveolar epithelium.

3.

Which cell type is located at the basal lamina of the olfactory mucosa?

- a. Basal cells***
- b. Brush cells***
- c. Olfactory cells***
- d. Sustentacular cells***
- e. All of the above***

Answer: a

Olfactory mucosa lines the roof and portions of the walls of the nasal cavity. It contains several cell types: basal cells, brush cells, olfactory cells and sustentacular cells.

Basal cells are located in the basal lamina. Brush cells are involved with general sensation of the olfactory mucosa. Olfactory cells are bipolar neurons that are the receptors for smell. Sustentacular cells are supporting cells. Sustentacular cells are most numerous cell type in the olfactory epithelium.

4.

Which cell type is most numerous in olfactory mucosa?

- a. Basal cells*
- b. Brush cells*
- c. Olfactory cells*
- d. Sustentacular cells*
- e. None of the above*

Answer: d

Olfactory mucosa lines the roof and portions of the walls of the nasal cavity. It contains several cell types: basal cells, brush cells, olfactory cells and sustentacular cells.

Basal cells are located in the basal lamina. Brush cells are involved with general sensation of the olfactory mucosa. Olfactory cells are bipolar neurons that are the receptors for smell. Sustentacular cells are supporting cells. Sustentacular cells are most numerous cell type in the olfactory epithelium.

5.

What type of epithelium is found in the respiratory mucosa of man?

- a. Non-ciliated pseudostratified columnar epithelium with goblet cells*
- b. Ciliated pseudostratified columnar epithelium with goblet cells*
- c. Simple columnar epithelium*
- d. Stratified squamous epithelium*
- e. Transitional epithelium*

Answer: b

In man, respiratory mucosa is composed of ciliated pseudostratified columnar epithelium with goblet cells. Respiratory mucosa contains several cell types: ciliated cells, goblet cells, basal cells, and brush cells. Respiratory mucosa is located lining the respiratory segment of the nasal cavity. It lines the conchae and the paranasal sinuses.

6.

What type of tissue makes up the "Adam's apple"?

- a. Compact bone*
- b. Spongy bone*
- c. Hyaline cartilage*
- d. Fibrocartilage*
- e. Elastic cartilage*

Answer: e

The "Adam's apple" is a nickname for part of the larynx formed by the thyroid cartilage. The thyroid cartilage is composed of hyaline cartilage.

7.

What type of tissue forms the alveoli in the lung?

- a. Simple squamous epithelium*
- b. Simple cuboidal epithelium*
- c. Simple columnar epithelium*
- d. Stratified squamous epithelium*
- e. Pseudostratified epithelium*

Answer: a

The alveoli are formed by simple squamous epithelium. Epithelium lines body cavities and surfaces. Simple squamous epithelium is "simple" because it is one cell thick. "Squamous" refers to the fact that the cells are flat.

8.

What type of tissue lines the paranasal sinuses?

- a. Simple squamous epithelium*
- b. Simple columnar epithelium*
- c. Stratified squamous epithelium*
- d. Ciliated pseudostratified epithelium with goblet cells*
- e. Transitional epithelium*

Answer: d

The paranasal sinuses are lined by ciliated pseudostratified epithelium with goblet cells.

9.

In which structure does gas exchange NOT occur?

- a. Alveolar duct*
- b. Alveoli*
- c. Alveolar sac*
- d. Respiratory bronchiole*
- e. Terminal bronchiole*

Answer: e

The respiratory tract is made of branching structures, much like the branches of a tree. The trachea branches into two bronchi which branch into smaller bronchi. The bronchi ultimately branch into smaller bronchioles. Bronchioles are distinguished from bronchi in that they do not have cartilage and submucosal glands. The terminal bronchioles are the last part of the airway in which gas exchange does not occur. Terminal bronchioles lead to the respiratory bronchioles. The respiratory bronchioles are the first section of the respiratory tree that gas exchange can occur. The alveoli duct is analogous to a thoroughfare with many cul-du-sacs branching off of it. At the end of the alveoli duct is an alveoli sac. An alveoli sac is a cluster of alveoli, much like a cluster of grapes. Alveoli are individual sacs where gas exchange occurs.

10.

Which cell secretes surfactant?

- a. Clara cell*
- b. Type I pneumocyte*
- c. Type II pneumocyte*
- d. Dust cell*
- e. Brush cell*

Answer: c

The Clara cell is found in the terminal bronchioles. The type I pneumocyte is a squamous epithelial cell. It covers most of the surface of the alveoli. The type II pneumocyte is also called a septal cell. The type II pneumocyte secretes surfactant. The dust cell is also called the alveolar phagocyte. Brush cells are occasionally, but rarely, seen in the alveolar epithelium.